MILITARY SPECIFICATION

SPRAYING UNIT, CLEANING-DEICING-DECONTAMINATING FLUID, TRUCK MOUNTED AF/S32M-1

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a truck-mounted, cleaning, deicing, and decontaminating fluid spraying unit, designated AF/S32M-1.

1.2 Classification. The spraying unit shall be of the following classes, as specified (see 6.2):

Class 1 - Basic spraying unit

Class 2 - Spraying unit with air blast system

Class 3 - Spraying unit with cleaning compound system

Class 4 - Spraying unit with air blast system and cleaning compound system.

2. APPLICABLE DOCUMENTS

*2.1 Government documents.

*2.1.1 Specification and standards. Unless otherwise specified (see 6.2.) the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-N-95	Naphtha, Aliphatic
VV-G-1690	Gasoline, Motor, Automotive, Leaded or Unleaded
PPP-T-60	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1730

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than 1-1/2 mils (22-1/2 pounds per ream) in thickness. The coating shall be

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DOD-D-1000	Drawing, Engineering and Associated Lists
MIL-S-3950	Switch, Toggle, Environmentally Sealed, General Specifications For
MIL-B-4792	Bumper, Rubber, Duplex Round
MIL-C-4851	Compressor, Air, Gas Turbine Driven, Trailer Mounted, 150 Air HP, Type MA-1A
MIL-H-5559	Hydraulic Fluid, Arresting Gear
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Specification For
MIL-A-8243	Anti-Icing and Deicing - Defrosting Fluid
MIL-A-8421	Air Transportability Requirements, General Specifications For
MIL-H-8790	Hose Assembly Rubber Hydraulic High Pressure (3,000 psi)
MIL-H-8795	Hose Assembly, Rubber, Hydraulic, Fuel, and Oil Resistant
MIL-E-22680	Engine, Gasoline, Commercial, Industrial Type, 10 Brake Horsepower and Above, General Specification
MIL-G-25013	Grease, Aircraft Ball and Roller Bearing
MIL-C-25769	Cleaning Compound, Aircraft Surface, Alkaline Waterbase
MIL-D-26124	Duct and Duct Assembly, Pneumatic, Flexible

STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standard For Service

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1188	Commercial Packaging of Supplies and Equipment.
MIL-STD-1230	Trucks and Truck-Tractors, Commercial Gasoline and Diesel Engine, 34,500 to 64,000 Pounds G.V.W., 6 x 4
MIL-STD-471	Maintainability Demonstration
MIL-STD-808	Finish, Protective and Codes For Finishing Schemes for Ground and Ground Support Equipment
MS3100	Connector, Receptacle, Electric Wall Mounting
MS3101	Connector, Plug, Electric, Cable Connecting
MS3106	Connector, Plug, Electric, Straight

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MS25331	Light Assembly, Press To Test Indicator
MS28782	Retainer, Packing, Back-Up, Teflon
MS35000	Battery, Storage, Lead-Acid, Waterproof
MS90347	Cable Assembly, External Electrical Power, Aircraft 28 Volt DC External Electric Power

Air Force-Navy Aeronautical

AN6227	Packing, "O" Ring Hydraulic

DRAWING

Air Force

54H6122	Winterization Components - Commercial Standardized, Air Force Vehicles

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The spraying unit shall consist of the following major components:

Description	See Requirement
a. Truck chassis	3.7.1
b. Spraying unit frame	3.7.2
c. Boom assembly	3.7.3
d. Work platform	3.7.4
e. Stabilizers	3.7.5
f. Hydraulic system	3.7.6
g. Auxiliary engine	3.7.7
h. Storage tanks	3.7.8
i. Pumping systems	3.7.9

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Description	See Requirement
j. Delivery systems	3.7.10
k. Compound system	3.7.11
l. Fluid heater system	3.7.12
m. Air blast system	3.7.13

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Protective treatment. When materials are used in the construction of the spraying unit that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.2 Dissimilar metals. Wherever possible, contact between dissimilar metals shall be avoided. Wherever such contacts are unavoidable, they shall be insulated in a manner approved by the procuring activity. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Standard steels. Steels in compliance with high commercial standards such as ASTM and AISI standards shall be acceptable for use in the spraying unit if the material is used in the manufacturer's standard commercial version of this equipment. Nonstandard steels may be used if a certified Mill test report is provided and approval of their use is granted by the procuring activity.

3.4.4 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.5 Thermal insulation material. All insulating material shall maintain its shape and not settle, sag, pack, nor disintegrate under conditions of vibration and temperatures encountered during the intended use.

3.5 Design and construction. The spraying unit shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

*3.5.1 Reliability. The spraying unit shall have a minimum MTBF (mean time between failure) of 102 hours with a minimum reliability of 0.97 for a 3-hour mission at a confidence factor of 0.90.

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*3.5.2 Maintainability. Maintainability requirements shall be in accordance with MIL-STD-471.

3.5.2.1 The spraying unit shall incorporate the principles of the paragraph entitled "Maintainability Principles" of MIL-STD-471 and the following:

a. Design and construction shall permit performing the maximum practicable number of necessary maintenance tasks using general purpose tools normally available commercially.

b. Replacement and adjustment of components and parts shall be possible with minimum disturbance of other parts or components.

3.5.3 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can only be installed in its proper operating position.

3.5.4 Design features. The spraying unit shall be designed as an airtransportable, self-contained pumping and spraying unit with an elevating work platform that is maneuverable through the range specified by controls on the platform and on the ground control panel, and mounted upon a 6-by-4 commercial truck chassis. The spraying unit shall be compact with all components arranged for maximum ease of operation and maintenance consistent with the utility and efficiency required to conform to the requirements specified herein.

3.5.5 Fluids. The spraying unit shall be designed to spray the fluids specified herein over external aircraft surfaces that are normally inaccessible. All components of the spraying unit shall be resistant to the following fluids where exposure to them is possible:

a. Water

b. Deicing and defrosting fluid conforming to MIL-A-8243.

c. Ethylene glycol conforming to MIL-H-5559.

d. Solutions containing cleaning compound conforming to MIL-C-25769.

3.5.6 Protective enclosures. Each component of the spraying unit subject to failure due to ice or precipitation shall be protected with a weatherproof cover.

3.5.7 Prevention of spillage. All fluid tanks and fluid-handling equipment shall be positioned, located, and baffled to prevent spillage of fluids over other parts of the spraying unit.

3.5.8 Drainage. Means shall be provided for complete drainage of fluid tanks, filters, plumbing, hoses, et cetera. Drains shall be located to be accessible from positions outside the spraying unit so that fluids will not be drained onto other parts of the equipment.

3.5.9 Safety. All mechanisms and parts designed and contructed to contain flammable liquids shall be positioned, located, and ventilated to preclude the possibility of fires or explosions caused by such liquids or vapors therefrom contacting heated surfaces or electrical apparatus. All rotating,

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folding, or reciprocating parts of such nature or so located as to become a hazard to operating personnel shall be fully enclosed or properly guarded.

3.5.10 Operating and service controls. All operating and service controls shall be readily operable by personnel wearing heavy, arctic gloves and clothing.

3.5.11 Mobility. The spraying unit shall be designed to withstand the stresses imposed when subjected to the mobility requirements specified herein. The spraying unit shall be designed for maximum stability during the mobile spraying operations specified herein with the elevating structure extended in any position.

3.5.11.1 Roadability. The spraying unit, with fuel and fluid tanks filled, shall withstand being driven over paved highways at 50 mph and over unimproved or graveled roads at 20 mph. The spraying unit shall be capable of propelling itself over unpaved surfaces around an airfield and over snow- and ice-covered airfields and adjacent areas.

3.5.11.2 Mobile operation. With the rear spring lockouts engaged and stabilizer jacks retracted, there shall be no evidence of instability or overturning of the spraying unit during mobile operation with a rated load of 500 pounds on the platform in any extended position. For such mobile operation, the speed of the spraying unit shall be limited to that of the lowest forward gear range of the truck. This mobile operation shall be limited to flat, paved areas such as parking ramps on an airfield.

3.5.12 Air transportability. The spraying unit shall be designed for air transportability in accordance with MIL-A-8421 in the assembled condition: i.e., with all components specified herein installed. The spray unit shall be designed and constructed to meet the requirements of MIL-A-8421 pertaining to floor loading, ramp negotiation, acceleration loads, tiedown provisions, and marking. The size and configuration of the spraying unit shall be within the maximum cargo dimensions for Type C-124 aircraft specified in table I. Restraint criterion shall be 3 gravitational units (g) forward, 2 g aft, 2g vertical, and 1.5 g lateral.

3.5.13 Bolts, nuts and pins. All bolts, nuts, pins, et cetera used in the construction of the spraying unit shall be cadmium plated and shall be furnished with Air Force standard locking devices. Safety wire, self-locking nuts, cotter pins, or lockwashers shall be used.

3.5.14 Wiring protection. Rubber grommets shall be provided to prevent chafing of cables where circuits pass through structural members.

3.5.15 Lubrication. The unit shall be designed and constructed to operate satisfactorily when lubricated with standard military lubricants specified in MIL-M-8090. A lubrication chart shall be provided directing attention to all lubricating fittings and shall specify the range and grade of lubricant required for critical temperatures. The chart shall specify grease conforming to MIL-G-25013 for all lubrication fittings on the air blast assembly that will be affected by the high temperature bleed air of the GTC. The chart shall be permanently attached to the unit in an accessible and convenient location.

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TABLE I. Cargo dimensions.

Length (Inches)	Width (Inches)			
	40	60	80	81-96
	Maximum Height (Inches)			
300	131	129	126	120
320	130	127	125	118
340	128	126	123	116
360	126	124	121	115
380	124	122	120	114
400	122	121	118	112

3.6 Performance. The spraying unit shall be capable of operating under and shall exhibit satisfactory resistance to the following:

a. Operating temperatures ranging from -65° to +130° F and storage temperatures ranging from -80° to +160° F.

b. Pressure altitude ranging from seal level to 6,000 ft.

c. A relative humidity of 95 to 100 percent at an operating temperature of 130° F and storage temperature of 160° F.

d. Self-induced vibration for all output conditions and vibration encountered during shipment and road operation.

e. All forms of precipitation encountered under field operation.

f. Windloads of at least 7 psf.

3.6.1 Operating positions. The spraying unit shall function satisfactorily and the elevating and rotating mechanisms shall hold position within the settling or creeping limits specified herein when the normal operating plane of the spraying unit is at an angle of 3° (or 5-1/4 percent) in any direction from the horizontal. The elevating and rotating structure shall be operable while the truck is in motion.

3.7 Details of components.

3.7.1 Truck chassis. The spraying unit, including all components and accessories, shall be mounted upon a truck chassis conforming to MIL-STD-1230 except as modified by the following requirements. All parts necessary to provide a complete and efficient vehicle shall be furnished in accordance with first-class commercial practice.

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3.7.1.1 Engine. The engine shall develop the necessary brake horsepower and torque required with the transmission and axle ratios used to allow the spraying unit to creep up a 20° (or 6 percent) inclined ramp at a maximum speed of 1-1/2 mph in both forward and reverse gears in the air-transportable weight condition.

3.7.1.2 Fuel tank. A side-mounted fuel tank with a minimum capacity of 60 gallons shall serve as the fuel reservoir for the truck engine, fluid heater, and auxiliary engine. Suction standpipes provided in the fuel tank shall allow for accumulation of at least 1 quart of water in the tank. A drain shall be located at the bottom of the tank for complete drainage of the fuel tank. The top of the tank shall incorporate the necessary fittings for attaching fuel lines to the various components.

3.7.1.3 Exhaust system. The tailpipe shall be located in such a manner that exhaust gases will neither enter the cab, impair the driver's vision forward of the cab, nor endanger an operator at the ground control panel of the spraying unit. If the exhaust pipe outlet is less than 18 inches above ground level, an efficient spark arrester shall be furnished as a part of the truck exhaust system.

3.7.1.4 Transmission. The truck shall be provided with either an automatic or semiautomatic transmission as described herein. The transmission shall provide for driver selection of individual gears or selection of ranges of gears required for efficient operation throughout the speeds specified under the mobility requirements. The shift control shall incorporate a positive indicator for each gear or range selected. The transmission shall be so protected that the operator cannot inadvertently shift into reverse. The transmission shall have a torque capacity adequate to transmit 110 percent of the maximum net engine torque throughout all reductions. A foot-actuated clutch member shall not be permitted. The transmission assembly shall be of a commercial design and capacity that has been produced and supplied as a part of a commercial vehicle.

3.7.1.4.1 Semiautomatic, interruption-drive transmission. When this type of transmission is furnished, it shall be one in which limited continuously variable torque ratio changes occur automatically, but in which further ratio changes are required and must be controlled by the operator with a consequent interruption of the drive.

3.7.1.4.2 Semiautomatic, continuous-drive transmission. When this type of transmission is furnished, it shall be either one in which limited continuously variable torque ratio changes occur automatically and in which further required ratio changes must be controlled by the operator with only instantaneous interruption of power flow, or one that affords only a limited number of torque ratio conversions that are affected automatically and with only instantaneous interruption of power flow.

3.7.1.4.3 Semiautomatic, continuous-drive, automatically stepped transmission. When this type of transmission is furnished, it shall be one in which a portion of the total torque conversion ratios is continuously variable, but in which the remaining ratio conversions are accomplished through range control by the use of automatically stepped gear ratios that do not require more than an instantaneous power-flow interruption with each selected range.

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3.7.1.4.4 Automatic transmission. When an automatic transmission is furnished, it shall be one in which all combinations of variable torque ratio changes and automatically stepped gear ratio changes can be accomplished by the operator with one setting of the shift control and without more than an instantaneous interruption of the power flow.

3.7.1.5 Tire tread. The tire tread shall be of the mud and snow type. A spare wheel or rim, tire, tube, and carrier shall not be furnished.

3.7.1.6 An aviation red flasher shall be so installed on the top center of the vehicle as not to interfere with other vehicle equipment. A control switch shall be located on the instrument panel for control of the flasher. Clearance lamps and reflectors shall be provided.

3.7.1.7 Stowage space. Weatherproof stowage space shall be provided of a size adequate for on-vehicle tools, tire chains and a highway warning kit. The space shall be accessibly located with provisions for being locked. Jack, jack handle, and wheel nut wrench shall not be provided.

3.7.1.8 Mud guards. Mud guards shall be provided to protect the mounted equipment from road splash from the rear wheels. (The flooring of the spraying unit may be adequate protection without the use of additional mud guards.) Sufficient clearance shall be provided between tires and fenders, mud guards, and mounted equipment to permit satisfactory installation and use of tire chains on the wheels.

3.7.1.9 Truck winterization. A winterization kit shall be installed on the truck prior to delivery. The kit shall provide standby-type winterization and shall include at least the following components: engine coolant heater (110V) cab heaters, oil-pan heater (110V), transmission heater (110V), frost shields, and winter front. Winterization shall be provided to insure operation at -65° F. Drawing 54H6122 may be used as a guide.

3.7.1.9.1 Receptacle, 110-V. The 110-V electrical components used for winterizing the truck and any 110-V heating units used for winterizing the auxiliary engine shall lead to one receptacle securely mounted and clearly labeled on the outside of the truck cab. The receptacle shall be of the male, three-prong, twist-lock type with a spring-loaded dust cap included as part of its housing. The third wire shall be adequately grounded to the truck.

3.7.2 Spraying unit frame. An integral supporting frame shall be provided for the part of the spraying unit that is mounted on the truck chassis. The frame shall be designed to allow mounting the spraying unit equipment on and operating it from any truck chassis frame conforming to the requirements specified herein. Large components such as the storage tanks and boom assembly shall be mounted directly to the supporting frame. The hydraulic stabilizers and spring lockouts shall be connected to the supporting frame.

3.7.2.1 Equipment flooring. A metal flooring shall be mounted directly on the supporting frame. The flooring shall serve as a means for supporting all components not mounted directly to the supporting frame, a work platform for servicing and maintaining the mounted equipment, and for preventing road splash from the rear wheels from reaching the mounted equipment. The flooring shall be of steel treadplate incorporating a nonskid walking surface. The flooring shall be of adequate thickness and shall have the necessary reinforcements and bracing to support the components fastened to it. The flooring shall be at least as wide as the rear wheels of the truck and shall be

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as long as necessary to support the mounted equipment. The flooring shall be designed so that fluids will drain to the outside. A metal ladder or rungs shall be provided on both sides of the spraying unit for an operator to climb onto the equipment flooring from the ground.

3.7.2.2 Ground control panel. A ground control panel shall be located on the driver's side of the spraying unit. It shall be mounted at a convenient height from the ground and in a vertical plane parallel to the longitudinal axis of the truck to facilitate observation and operation of the instruments and controls by an operator standing on the ground. The control panel shall be constructed of sheet steel No. 12 (0.1046 inch) U.S. gage or thicker, and supported by a welded-steel, self-supporting framework. The panel shall be rigidly fastened to the equipment flooring. All valves, indicators, gages, switches, and controls mounted on the control panel shall be arranged to avoid confusion of the operator and shall be identified as to function and use by permanently etched or engraved metal or plastic nameplates permanently attached to the control panel. All instruments and gages mounted on the control panel shall be uniform in appearance. Instruments subject to error due to vibration shall be shock mounted. The back of the control panel shall be enclosed with a removable, metal, weather-resistant cover to provide maximum protection to gages, wiring, electrical connections, and terminals.

3.7.3 Boom assembly. A boom assembly consisting of two hydraulically-operated booms hinged to a rotating turntable shall support the work platform. The structure shall be designed for a working load of 500 pounds on the platform during mobile operation as specified in 3.5.11.2 and 750 pounds on the platform while in any extended position with the stabilizer jacks extended. A 1,500-pound load placed on the platform when the boom assembly is in any extended position and the stabilizer jacks extended shall not cause instability or overturning of the spraying unit. The booms supporting structure shall be designed with a safety factor of at least 3 in all positions. A 2,250-pound load exerted at the boom platform hinge with the boom in any extended position shall not cause permanent deformation of the booms, turntable, or any component part.

3.7.3.1 Range of platform movement. The boom assembly shall be capable of unlimited rotation in either direction in the horizontal plane. The booms shall raise the platform to a maximum height of at least 48 feet (measured to the platform floor) above the ground with an overhang of at least 3 feet from the rear and both sides of the spraying unit. The length and configuration of the booms shall allow the platform to be moved along a horizontal surface 20 feet above the ground until an overhang of at least 22 feet from the rear and from both sides of the spraying unit is obtained. The booms shall be designed to allow the platform to be lowered to within 12 inches (measured to the platform floor) of the ground to the rear and to both sides of the spraying unit. For normal storage, travel, and shipment of the unit, the booms shall fold to essentially horizontal positions in a vertical plane parallel to the longitudinal axis of the spraying unit. The platform shall be suspended to the rear of the unit with the booms folded. The boom assembly shall be designed to allow the platform to be lowered from the folded (travel) position for mounting from the ground. The range of platform movement is shown on figure 1.

3.7.3.2 Boom operation. The upper boom, lower boom, and rotating turntable of the boom assembly shall be hydraulically actuated by separate mechanisms. The actuating mechanisms shall control the raising, lowering, and rotating movements of the boom assembly. It shall be possible to operate the booms and

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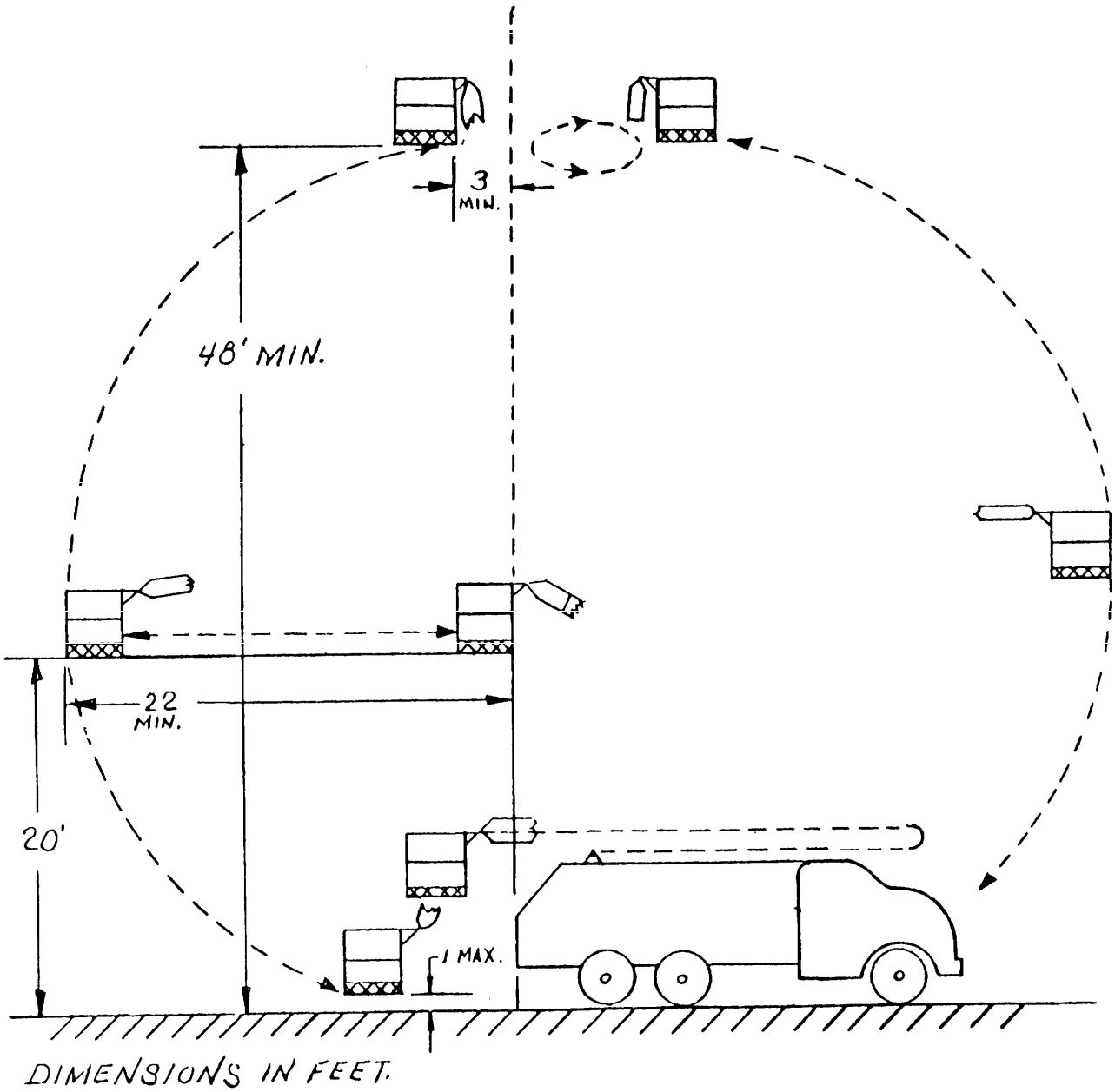


Figure 1. Range of Platform Movement

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turntable independently or in any combination of raising, lowering, and rotating the boom assembly at the same time. Movement of the boom assembly shall be controllable from the platform and from the ground control panel. Boom controls on the ground control panels shall duplicate those on the platform and shall override or cancel the controls on the platform.

3.7.3.3 Boom construction. The booms shall be constructed of full length stressed skin (over stiffeners), I-beam, boxed-channel, or tubular-type members of minimum size and weight consistent with the requirements herein. Fluid lines and other parts that are fastened to the booms shall be positioned for easy inspection and replacement and maximum protection against damage afforded by the configuration of the boom. The bombs shall be hinged together, to the platform, and to the turntable by replaceable pins or shafts. Renewable surfaces and lubricating fittings shall be provided for bearing surfaces subject to wear under continued use.

3.7.3.3.1 Tiedown device. A means shall be provided to tie down the boom assembly to prevent unactuated raising of the booms during shipment of the spraying unit. The tiedown device shall be an integral part of the spraying unit to prevent inadvertent loss during use.

3.7.3.3.2 Boom support. Saddles shall be provided to support the upper boom on the lower boom and to support the forward end of the boom assembly. The saddles shall be designed to prevent side motion of the booms. The design and positions of the saddles shall be consistent with the type of boom construction used and shall not cause damage of the booms during use and transit.

3.7.3.4 Turntable. The boom assembly shall be rotated by a hydraulically-operated motor. The turntable shall be constructed to eliminate all side thrust on the motor shaft. The unactuated movement or play in the turntable construction and operation shall be held to a minimum and shall not permit the horizontal platform creep to exceed 2 inches per hour with a 750-pound load on the platform. The motor shall rotate the loaded platform 360° in not less than 45 nor more than 60 seconds.

3.7.3.4.1 Turntable locking device. A mechanical device shall be provided that, when engaged, will prevent rotary movement of the turntable with the booms folded in the traveling position. The locking device shall be designed to withstand all shock loads encountered during shipment and movement of the spraying unit. When engaged, there shall be no strain on the rotating gears or hydraulic motor due to shifting action of the boom assembly. Indicators shall be provided and shall be so located as to be visible to the operator at the ground control panel to assist in aligning the locking device.

3.7.3.4.2 Swivel fittings. Swivel fittings shall be provided for all fluid and electrical lines between the rotating and stationary parts of the turntable to permit unlimited rotation of the boom assembly.

3.7.4 Work platform. A work platform shall be hinged to the end of the upper boom. The platform shall be suspended from the boom with the hinge joint at the upper rear of the platform structure. The platform shall be of sufficient area to permit complete freedom of movement of two persons while spraying fluid, performing inspection and maintenance operations, and operating the platform controls. The minimum dimensions of the working area shall be 36 by 48 inches. The platform shall be designed for a normal working load of 750 pounds and shall support the loads specified in 3.7.3. All parts of the

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platform that contact the aircraft, under normal operation and in case of equipment failure, shall be padded with a resilient, replaceable bumper. The bumper material shall be of half-round, three-quarter round, full-round, or double-full round sponge rubber core covered with a woven cotton fabric in accordance with applicable portions of MIL-B-4792. Projects or sharp corners shall be rounded off and thickly padded. A lifting ring shall be provided for connecting the lifting strap specified in 3.7.13.9.

3.7.4.1 Guardrails. Two rigid guardrails shall be provided around the perimeter of the work platform. The upper guardrail shall be 36 to 40 inches above the work platform floor and shall be of one piece with no gateway. The lower guardrail shall be midway between the kickplate and upper guardrail. The lower guardrail on each side of the work platform shall be hinged to provide access gateways to the platform. The gateways shall be at least 30 inches wide. A positive safety catch shall be provided on the gateway rail to hold the gate closed. The safety catch shall be operable by a person wearing heavy, arctic gloves. All joints in the guardrail shall be welded except for the gateways. The guardrails shall support a 500-pound load in any direction without permanent deformation of any member.

3.7.4.2 Platform flooring. The work platform shall have a reinforced expanded metal flooring with a clear working area of at least 36 inches from front to back and at least 48 inches wide. The expanded metal flooring shall have diamond-shaped openings a maximum of 3/4 inch wide and shall not weigh more than 2 psf. The upper surface of the expanded metal shall have projections to provide safe footing to personnel wearing heavy arctic boots. To eliminate sharp prongs at the perimeter of the flooring, the expanded metal shall be cut or sheared on the centerline of the diamonds. The expanded metal shall be arc welded to the frame with the welds placed at intervals of not more than 6 inches on the inside of the mesh toward the center of the platform. No point on the flooring shall deflect more than 1/4 inch when subjected to a concentrated vertical load of 300 pounds covering an area of 25 square inches. This loading shall not cause permanent deformation of the flooring.

3.7.4.3 Kickplate. A kickplate, 4 to 6 inches high, shall be provided around the perimeter of the work platform flooring.

3.7.4.4 Platform leveling system. Two systems shall be provided for keeping the platform horizontal at any boom position. Each system shall be independently capable of keeping the platform level and shall be designed with a safety factor of not less than 4. The leveling system shall be designed with minimum stretch due to loading the platform. If the leveling system stretches, an adjustable link shall be provided and the platform shall be adjusted to be level under load. A stranded-cable leveling system shall not be acceptable.

3.7.4.5 Platform controls. Hand-operated, hydraulic control valves for each boom and the turntable shall be provided on the platform structure and shall be easily operable by a person wearing arctic clothing. If necessary, the control valve handles shall be lengthened to improve the fine metering control of the hydaulic fluid. The control handles shall be on the front of the platform structure (away from the truck) and shall not extend above the top guardrail. The controls shall be so positioned that the operator can actuate the handles by pushing them away from him or pulling them towards him in a horizontal direction. The control valves shall have a self-centering feature and a definite feel of the neutral position. Smooth operation of the

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boom assembly shall be possible with the control valves used. The controls shall be plainly and permanently marked as to function and use. Provisions shall be made to prevent inadvertent actuation of the controls by an operator performing maintenance on an aircraft from the work platform.

3.7.4.6 Safety belts. Two positive-lock-type safety belts with quick-release features and a design strength of 500 pounds each shall be provided on the work platform. The belts shall be designed and positioned to allow freedom of movement of the operators on the platform.

3.7.4.7 Floodlights. Two 50-cp, sealed-beam floodlights shall be mounted on the platform and adjustable to shed light on the front, side, above, and below the platform for night operation. The floodlights shall be protected to prevent inadvertent breakage and interference with the aircraft during movement of the platform. The floodlights shall operate from the 24-V electrical system of the auxiliary engine. On-off switches shall be provided on each floodlight assembly at the platform. A switch shall be provided on the ground control panel in series with the switches on the platform to cut off electricity to the platform.

3.7.4.8 Signaling system. A loud signaling device, such as a large truck horn, shall be provided. It shall be sounded by pushbuttons on the work platform and the ground control panel. The signal system shall provide a two-way communication between operating personnel on the platform and by the ground control panel.

3.7.5 Stabilizers. Hydraulically-operated stabilizing jacks or outriggers shall insure adequate stability with relation to the ground while operating the platform under maximum working loads. Necessary controls for extending, adjusting, retracting, and locking each stabilizing jack shall be located on the ground control panel. The stabilizing jacks shall be equipped with jack base plates having sufficient area to prevent settling when in contact with soil having a load bearing capacity of 5 tons per square foot. The jacks shall be designed to retract to within the overall dimensions specified herein for storage, traveling, and mobile operation.

3.7.5.1 Rear spring lockouts. Rear spring lockouts shall be provided for mobile operation of the spraying unit as specified in 3.5.11.2. The spring lockouts shall be removable and replaceable so the spraying unit and components can be transferred to another truck chassis. The lockouts shall be hydraulically operated. Necessary controls for engaging, locking, and disengaging the lockouts shall be located on the ground control panel.

3.7.6 Hydraulic system. A complete hydraulic fluid system shall be provided consisting of the components required to perform all hydraulically powered operations of the boom assembly and stabilizers with the degree of control specified herein. The hydraulic system shall be designed for use with hydraulic oil conforming to MIL-H-5606 and shall be furnished filled with the proper amount of that oil. The hydraulic system shall be designed and installed in accordance with best commercial practices as specified herein.

3.7.6.1 Hydraulic cylinders. All hydraulic cylinders used to actuate the booms and the stabilizers shall incorporate seals to prevent entrance of moisture into the cylinders and leaks of hydraulic fluid from the cylinders. O-type rubber packings in accordance with AN6227 shall be used in the cylinders where static seals are required. O-rings protected by teflon backup rings conforming to MS28762 shall be used where reciprocating movement is

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encountered. The hydraulic cylinders for actuating the booms shall be provided with check valves to prevent descent of the booms in case of hydraulic line failure. The hydraulic seals in the system shall not permit the unactuated vertical settling of the loaded platform to exceed 2 inches per hour when the booms are elevated to form an angle of 45° with the horizontal plane.

3.7.6.2 Hydraulic lines and fittings. Nonflexible metal tubing shall be used in the hydraulic system where flexing is not encountered during operation of the boom assembly. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, best visual inspection for failures, and maximum protection from damage during operation of the boom assembly. Flexible hose shall be used where flexing of hydraulic lines is encountered and where the installation of components requires a flexible coupling. Hydraulic hose assemblies shall conform to MIL-H-8790 and MIL-H-8795, depending upon hydraulic system pressure requirements. Hose assembly installations shall be made consistent with the minimum bend radii specified for the size hose used. Standard AN straight-thread, flared-type fittings shall be used throughout the hydraulic system. Wherever special fittings and joints requiring standard pipe threads are used, adaptation to flared-type fittings shall be made as appropriate. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic lines when necessary.

3.7.6.3 Hydraulic pump system. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulically-powered components as specified herein shall be provided. The pump while operating at normal rated speed, shall have sufficient capacity to simultaneously elevate the platform (under maximum rated load conditions) from the lowest to the highest position and to rotate the platform through 360° in not less than 45 nor more than 60 seconds. The hydraulic pump shall be driven by the auxiliary engine through a flexible coupling or multibelt drive. Chain drives shall not be used. The pumping system shall include a surge suppressor and an adjustable pressure regulator with the range of adjustment to allow raising of any working or test loads on the platform as specified herein. A hydraulic pressure gage shall be mounted on the ground control panel.

3.7.6.4 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided with a capacity to hold the maximum volume of fluid that will be retained in the reservoir under any position of the boom assembly plus a minimum of 10 percent additional air space to handle expansion, foaming, and surging of the fluid in the reservoir. The reservoir shall have a filler opening of at least 1-inch diameter located to facilitate filling the tank from a 1-gallon container using a funnel. A vented filler cap shall prevent oil from splashing out of the tank during movement of the spraying unit. The cap shall be securely attached to the tank by a chain to prevent loss. An oil strainer that can be easily removed for cleaning shall be included at the outlet of the reservoir. Provisions shall be made for draining the reservoir. The drain shall be so located the oil will not drain on other components of the spraying unit and to allow recovery of the oil in a container. A bayonet or dip stick shall be provided to determine the oil level in the reservoir. The reservoir shall be marked to show use of oil conforming to MIL-H-5606 and the maximum oil level in the reservoir with the booms folded and the lines, cylinders, et cetera, full of oil.

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3.7.6.5 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within reasonable limits during continuous operation of the unit and use of the platform for long periods of time as a maintenance platform. Means shall be provided to prevent excessive heating of the fluid such as by unloading the system pressure during standing operations. Fluid temperature changes (other than those due to change in ambient temperature) that contribute to or cause the creep or settling limits specified herein to be exceeded shall be eliminated. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in fluid temperature.

3.7.7 Auxiliary engine. A gasoline engine conforming to group II of MIL-E-22680, and as specified herein, shall be provided to drive all operating components of the spraying unit other than the truck.

3.7.7.1 Capacity. The engine capacity shall be such that the continuous-duty horsepower input required by all items driven by the engine will not exceed 75 percent of the full-throttle rating of the engine at sea level and at 60°F.

3.7.7.2 Cylinders. The engine shall have not less than four cylinders.

3.7.7.3 Environmental requirements. The engine shall be capable of starting and operating as specified in MIL-E-22680 under the environmental operating conditions specified for the spraying unit in 3.6.

3.7.7.4 Gasoline engine accessories. The following requirements shall be met in addition to or in lieu of those specified in MIL-E-22680 for engine accessories and equipment.

3.7.7.4.1 Fuel system. The necessary fuel lines and a positive displacement-type fuel pump shall be provided to deliver fuel from the truck fuel tank to the auxiliary engine at the maximum rate of fuel consumption. A fuel shutoff valve between the fuel tank and sediment bowl shall prevent loss of fuel from the entire line during cleaning of the sediment bowl.

3.7.7.4.2 Exhaust system. The engine exhaust system shall include an efficient spark arrester.

3.7.7.4.3 Governor. A variable-speed governor shall automatically maintain the selected operating speed of the engine during operation of the spraying unit. A manually-operated throttle control shall be provided that can be locked in any position from the manufacturer's recommended idling speed to maximum operating speed.

3.7.7.4.4 Cranking system. An electric cranking system shall be furnished with a means provided for manually cranking the engine in case of electrical failure. The electric cranking system shall crank the engine at a speed sufficient for starting at -65° F with the winterization system operating.

3.7.7.4.5 Electrical system. The 24-V electrical system of the auxiliary engine shall be grounded to be consistent with the grounding of the electrical system of the truck. Two MS35000-4, 12-V batteries shall be provided. Grommets or clamps shall be provided to prevent chafing of cables and wiring where they pass through structural members.

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3.7.7.4.6 Storage battery insulation. The storage batteries shall be protected as necessary to prevent freezing of the electrolyte at -65° F.

3.7.7.4.7 Generator. The battery-charging generator shall have a minimum rating of 60 amp. The generator rating shall be equal to or greater than the maximum continuous-current requirement during operation of all electrical components powered by the auxiliary engine electrical system.

3.7.7.4.8 Clutch. The engine shall be equipped with a hand-operated friction clutch to allow for starting the engine under no load. The clutch lever shall be accessible to the operator at the engine control panel.

3.7.7.4.9 Winterization. The engine shall be winterized for operation at -65° F. The necessary winterization components shall be provided and installed. Standby heating provisions such as headbolt and oilpan immersion heaters may be 100V. If the heaters are required during operation of the engine to maintain desired temperatures, they may operate on the 24-V electrical system of the engine. All 110-V components shall be connected to the receptacle on the truck as specified in 3.7.1.9.1.

3.7.7.4.10. Engine controls and instrumentation. All engine controls shall be mounted on an engine control panel provided near the ground control panel of the spraying unit and shall be readily accessible to an operator standing on the ground at the left side of the unit. All controls shall be designed to facilitate ease of operation by personnel wearing heavy arctic clothing. Engine instruments as specified in MIL-E-22680 shall be mounted on the engine control panel. In addition a tachometer-hourmeter indicator, a head temperature gage, and an oil temperature gage shall be located on the engine control panel.

3.7.8 Storage tanks. Metal storage tanks, with capacities as specified herein, shall be provided as integral parts of the spraying unit to serve as reservoirs for the fluids to be sprayed. The tanks shall be baffled, as necessary, to prevent excessive sloshing of the fluid. The tanks shall be resistant to corrosion and deterioration caused by the fluids. Each tank shall be provided with a dial-type gage graduated in not greater than 25-gallon increments to indicate the quantity of fluid therein. A vent shall be provided at the top of each fluid tank. Each tank shall be equipped with a drain plug. A strainer shall be provided at the outlet of each tank to prevent suspended solids from entering the plumbing system where they might harm pumps or clog filters, lines, or nozzles. The strainer shall be readily removable from the line to permit periodic cleaning. A filler opening shall be provided in the top of each tank. Spring loaded or counterbalanced quick-opening hinged filler caps shall be provided on each tank. Metal steps or rungs shall be provided for personnel to reach the filler openings in the top of the tanks.

3.7.8.1 Main fluid tanks. Two 500-gallon storage tanks shall be provided as reservoirs for water, deicing fluids, and decontaminating fluids used for the various operations performed with the spraying unit. The tanks shall be constructed of stainless steel of an appropriate grade and thickness.

3.7.8.1.1 Tank enclosures. The two 500-gallon tanks shall be enclosed by a removable metal cover. The minimum clearance between the tank and its metal cover shall not be less than 2 inches. Each tank shall be completely enclosed except for access openings for filling and servicing, draining, and making necessary pipe connections to the tanks, and for installation of the

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liquid level gages in the tanks. The tank enclosure shall be designed to prevent liquids from collecting within the enclosure. The space between the tanks and between the tanks and covers shall be insulated to a heat transfer coefficient (K-factor) of less than 0.20.

3.7.8.1.2 Main tank opening. A manhole having a diameter of not less than 20 inches shall be located in the top of each 500-gallon tank. The manholes shall have covers, firmly clamped, and shall be used as a means of access at the time of overhaul and inspection. Each manhole shall be equipped with a gasket to insure a seal between the tank and the cover.

3.7.8.1.3 Fluid return line. The fluid return lines to the main tanks shall distribute the fluid across the bottom of the tank to provide the best circulation for heating and mixing of the fluid in the tank by pumping through the fluid heater and back to the tank.

3.7.9 Pumping systems. Pumping systems shall be provided with the necessary components to deliver the fluids specified herein at the flows, pressures, and temperatures specified to accomplish the required spraying operations. Provisions shall be made for complete drainage of fluids from all components of the plumbing systems. Plumbing for the pumping systems shall conform to figure 2 and shall be properly installed to allow for easy maintenance and replacement of component parts.

3.7.9.1 Pump ratings. Pumps shall be efficient and dependable in design, capable of handling the respective fluids, and of adequate rating to deliver fluids as specified herein under all operating conditions.

3.7.9.2 Pump drives. Pumps shall be driven by the auxiliary engine through flexible couplings or multibelt drives. Chain drives shall not be used. When a compound system is specified, a clutch shall be provided to disengage power to the compound pump. When use, unloading valves shall be of the multiposition selector type.

3.7.9.2.1 Pump drive adjustment. Where necessary, flexible lines and adjustable mounts shall be provided to maintain pump operation. Hose assemblies used in the flexible couplings shall be consistent with the type used for the delivery hoses specified in 3.7.10.

3.7.9.3 Main pump. The main pump, suitable for use with flammable fluids, shall be provided for delivery of fluids from the storage tanks to the spray nozzles. The pump capacity shall be sufficient to deliver not less than 50 gpm at working pressures up to 150 psig measured at the spray nozzles. The main pump shall also deliver decontaminating fluids at the specified flow and pressures for a continuous period of 100 hours without a reduction in performance greater than 10 percent.

3.7.9.3.1 Pump suction. The main pump shall lift water 15 feet while delivering rated output through the rinse system under all of the conditions specified herein. A two-position selector valve shall be provided in the suction line of the main pump to direct fluids from the tanks or the suction hose.

3.7.9.3.2 Pressure adjustment. The fluid pressure of the main pump shall be such that it can be adjusted to and fixed at any point within the range of 50 psig to maximum operating pressure. A pressure gage reading from 0 to 300 psig shall be mounted on the ground control panel for measuring the fluid pressure at the main pump outlet. A means shall be provided for bypassing the

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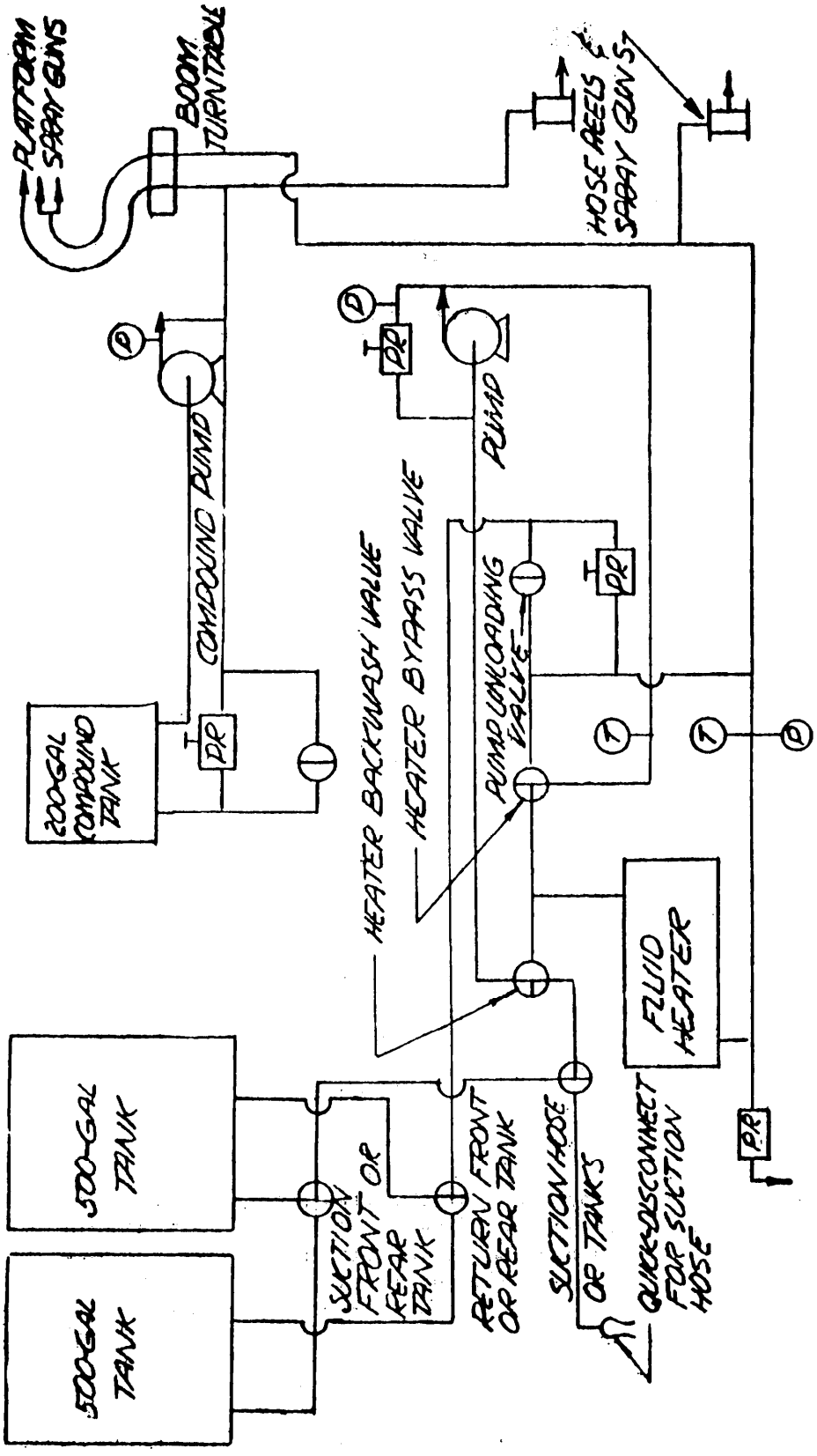


Figure 2. Typical Pumping & Spraying System Flow Diagram

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pressure regulator to unload the pump such as during filling and recirculating operations. Surge suppressor, relief valves, bypass valves, and return lines shall be included to insure an even flow of fluid without surging or fluctuation in pressure.

3.7.9.3.3 Main pump plumbing. Necessary plumbing shall be provided to convey the fluids from either 500-gallon storage tank and to permit filling either tank by the pump. The pump shall deliver the fluid to and through the fluid heater, to the nozzles, or back to the tank. The pressure regulator and unloader valve shall be downstream of the fluid heater. A means shall be provided to bypass the fluid heater. The hand lines on the hose reels or on the platform shall be capable of spraying fluids delivered by the main pump from the storage tanks or directly through the suction system. Flow control valves and regulators shall be clearly marked as to their function and shall be readily accessible to the operator standing by the ground control panel. The high-pressure plumbing shall be adequate to deliver 50 gpm to the work platform at pressures up to 150 psig.

3.7.10 Delivery systems. The delivery system for the main pump shall be designed to operate independently and concurrently with the compound spray system. All components of the delivery system shall withstand the working pressures specified and a hydrostatic pressure of at least twice the specified working pressure without leakage or failure.

3.7.10.1 Hoses. Low-temperature-resistant hoses having oil- and grease-resistant outer covers and possessing properties to resist the fluids being handled shall be provided on the spraying unit. Hoses shall be provided for the main pumping system, suction system, and for the compound system when it is provided. Hoses shall be provided on the work platform and on hose reels mounted on the spraying unit for spraying operations at ground level. The hoses shall be of the number, length, and type specified herein, complete with all necessary couplings to connect the spray guns to the pumping systems and to connect the lengths of hose.

3.7.10.1.1 High-pressure hoses. One 100-foot hose, mounted on a hose reel, shall be provided to deliver fluid from the main pumping system for use of operators on the ground. The hose may consist of two 50-foot lengths coupled together. Two hoses, each at least 8 feet long, shall be provided on the work platform to connect the spray guns to the high-pressure spray fluid line on the upper boom. The hoses shall withstand the ambient and fluid temperatures and the pressures encountered during all operating conditions specified herein. The high-pressure hose assemblies shall conform to MIL-H-8795.

3.7.10.1.2 Suction hose. One 15-foot length of suction hose shall be provided for filling fluid tanks by means of the main pumping system. The hose shall be capable of carrying a suction flow of at least 50 gpm and shall be constructed to withstand a vacuum of 20 inches Hg without showing signs of collapse. The hose shall be equipped with a quick-disconnected-type fitting for attaching the hose to the suction side of the main pump. The hose shall be flexible at -65° F. A strainer and a metal tube equipped with a foot check valve shall be provided. The strainer and the metal tube shall be equipped with quick-disconnect type fittings for attaching to the inlet end of the suction hose. It shall be possible to insert the metal tube and check valve through the 2-inch diameter opening in a 55-gallon metal drum and shall extend to the bottom of the drum.

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MIL-M-8090 A 1 b injection shall be provided directing attention to all

3.7.10.2 Hose reel. A hose reel shall be located on the rear side of the spraying unit for storing the 100-foot main delivery hose. A shutoff and drain valve shall be provided to permit draining and removal of the hose and hose reel without interfering with operation of the rest of the system. The hose reel shall be enclosed with access doors that will not interfere with reeling or unreeling of the hose. Hose roller guides shall be provided to aid in reeling and unreeling the hose. The hose reel shall be equipped with swivel joints to allow the hose to deliver fluids as specified while connected to the reel with any portion of the hose unreeled. The reel shall be equipped with an electrically powered means of rewinding the hose. The rewind switch shall be adjacent to the hose reel and protected from interference with the hose. The reel shall be equipped with a manual rewind handcrank and brake. The diameter of the hose reel shall be consistent with the bend radii specified for hose in MIL-H-8795.

3.7.10.2.1 Suction hose storage. Provisions shall be made on the spraying unit for storage of the suction hose when not in use. The stored hose shall be accessible from ground level.

3.7.10.3 Spray guns and nozzles. Spray guns and nozzles shall be provided for the three spray hoses, two on the platform and one on the hose reel. The spray guns shall be attached directly to the delivery hoses and the nozzles shall be attached to the guns. The connection between the delivery hose and the spray gun shall allow the gun to swivel on the hose. A means of storing and protecting the spray guns and nozzles when not in use shall be provided on the spraying unit. The spray-gun racks shall readily hold the spray guns while attached to the delivery hoses. Storage provisions for the spray guns on the platform shall not interfere with freedom of movement of operating personnel on the platform. The spray guns shall be operable by personnel wearing heavy arctic gloves.

3.7.10.3.1 High-pressure spray guns. Three spray gun assemblies, one for each hose, shall be provided to control the flow through the high-pressure hoses. The spray gun body shall be of lightweight material and shall be approximately 18 inches long with 2 handgrips provided. The outer end of the spray gun body shall be threaded for connection to interchangeable nozzles. The spray guns shall be adjustable to deliver a widely spread fine mist, a heavy conical spray, or a solid straight stream. Delivery of the fluids shall be positively controlled by a trigger-type valve at the handgrip. The spray guns shall incorporate insulated or ventilated handgrips to protect the operator against heat when spraying heated fluids. A metal guard shall be provided on the outer end of the spray gun body to prevent the fluid running down on the operator's hands. The spray gun shall deliver not less than 15 gpm measured at the gun and shall withstand and operate properly under working pressures up to 150 psig.

3.7.10.3.2 High-pressure nozzles. Lightweight nozzles shall be furnished for each of the spray guns specified in 3.7.10.3.1. The nozzles shall be of the replaceable orifice disc type. The nozzle shall be furnished with various sized orifice discs to permit a flow of at least 5, 10, and 15 gpm at working pressure of 100 to 150 psig.

3.7.11 Compound system. When specified (see 6.2), a cleaning compound system shall be provided on the spraying unit. A 200-gallon tank shall be furnished for holding cleaning compound conforming to MIL-C-25769. The tank shall be fabricated from sheet steel having a minimum thickness of 0.10 inch.

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3.7.1 Truck chassis. The spraying unit, including all components and

3.7.11.1 Compound tank opening. An opening of at least 12 inches in diameter or width shall be provided in the top of the tank for filling, cleaning, and maintaining the tank.

3.7.11.2 Cleaning compound pump. A pump shall be provided to deliver cleaning compound conforming to MIL-C-25769 at a minimum rate of 10 gpm at a working pressure 50 ± 5 psig measured at the nozzles.

3.7.11.2.1 Compound pump plumbing. The pumping system shall include the necessary plumbing to allow the fluid to be delivered from the tank through the handlines on the ground and on the work platform or to be circulated through the pump and back to the tank for purposes of mixing. The plumbing shall include a filter, pressure regulator, bypass line, and flow-control valves. The system shall be separate from the other pumping system. A pressure gage shall be mounted on the ground control panel to indicate the fluid pressure at the pump outlet.

3.7.11.3 Compound hoses. One 100-foot hose, mounted on a hose reel, shall be provided to deliver fluids from the compound pumping system to the compound spray gun for use of an operator on the ground. The hose may consist of two 50-foot lengths coupled together. One hose, at least 8 feet long, shall be provided on the work platform to connect the compound spray gun to the compound line on the upper boom. Each hose shall be operable independently or concurrently with the others. The compound delivery hose assembly shall be in accordance with MIL-H-8795. The compound hose reel shall meet the requirements of 3.7.10.2.

3.7.11.4 Cleaning compound spray guns and nozzles. Spray gun assemblies and nozzles shall be provided on the compound hoses to deliver the cleaning compound fluids at the rate of at least 5 gpm at a working pressure of 50 ± 5 psig. Each spray gun shall be of lightweight material with a 5-foot extension attached. Delivery of the compound spray shall be positively controlled by a trigger-type valve at the handgrip. A metal guard shall be provided on the gun to prevent the compound from running down on the operator's hands. A nozzle shall be furnished for the end of the gun extension that will deliver the compound in the form of a gentle, conical spray.

3.7.12 Fluid heater system. A fluid heater system consisting of a gasoline-burning heater and all accessories shall be provided. The fluid heater system shall have a minimum heat output capacity of 1,000,000 BTU per hr to be generated at an overall thermal efficiency of at least 75 percent under all operating conditions specified herein. The heater system shall be designed for fluid flows of at least 50 gpm. All components of the heater system shall be designed to operate satisfactorily under all environmental conditions specified herein.

3.7.12.1 Heating coil. The gasoline-burning fluid heater shall be of the progressive heating, continuous tubular coil-type containing no drums or extended pressure chambers. The heating coil shall be of welded steel construction and shall withstand a hydrostatic test pressure of 1,000 psig. The coil shall be designed and positioned to permit easy removal for repair and replacement. The coil shall be designed to prevent the formation or condensation of excessive carbon deposits on any part of the coil.

3.7.12.1.1 Safety valve. A safety valve shall automatically relieve the pressure in the heater coil when the normal operating pressure is dangerously

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exceeded. The discharge from a safety pop-off-type relief valve shall be directed vertically towards the ground.

3.7.12.1.2 Descaling fittings. Valves and fittings shall be provided on the intake and outlet sides of the heating coil as a means of permitting forward or reverse descaling by introduction of a solution of muriatic acid containing 2 percent (by weight) of paraformaldehyde as an inhibitor for the scale removing operation. The scale removing acid shall be circulated through the heater system by the fluid pump without resorting to any auxiliary pumping means.

3.7.12.2 Burner. The burner shall be of the pressure-atomizing type equipped with fully automatic ignition and controls for smokeless operation under all conditions including starting and stopping of the heater. It shall be designed for burning gasoline conforming to VV-G-1690.

3.7.12.2.1 Burner dependability. The burner shall not be affected during operation by vibration, sudden shock, or tilted operation.

3.7.12.2.2 Burner ignition. A complete, high-tension ignition system shall be provided that will give dependable ignition at temperatures as low as -65° F. The ignition system of the burner shall be wired into the safety control circuit to prevent any possible momentary discharge of fuel until perfect ignition is assured, and to protect against explosion or damage caused by such failures as weak ignition spark, cracked ignition electrode, electric current failure, or any other factor that may contribute to the malfunction of the burner. The ignition system shall incorporate fail-safe features throughout. If a vibrator is used, it shall have a two-position switch to permit immediate shifting to reverse set of vibrator points integral with the ignition system in the event of failure of the first set of points. The ignition assembly shall be designed to facilitate replacement of any vibrator used.

3.7.12.2.3 Burner mounting. The burner and forced-air system shall be so designed that access openings, where flame or products of combustion might escape, are a minimum of 34 inches above ground level to minimize the danger of explosion of surrounding fuel vapors. The burner shall be mounted to facilitate cleaning and replacement.

3.7.12.3 Combustion air blower. The combustion air blower shall supply all requisite air at sufficient pressure to evacuate all combustion gases without the aid of any induced draft blower. The blower shall continuously purge the combustion chamber while the heater is in operation. The blower shall be electrically driven and shall operate on the main heater switch. A means shall be provided to prevent re-establishment of burner ignition until the combustion air blower has had sufficient time to purge the combustion chamber and the heater exhaust chambers.

3.7.12.4 Temperature controls. The temperature of the fluid being heated shall be controlled by a cycling switch in direct contact with the fluid on the outlet side of the heater coils. The cutout temperature of the cycling switch shall be 195° $\pm$ 5° F and the cut-in temperature shall be 180° $\pm$ 5° F. A high-limit switch shall be provided at the fluid outlet with a cutout temperature of 210° $\pm$ 5° F with automatic reset to restart heater operation at a fluid temperature of 10° to 20° lower than the cutout temperature.

3.7.12.5 Heater fuel system. A complete fuel system shall be provided as an integral part of the fluid heater system.

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3.7.12.5.1 Fuel pump. The fuel pump shall be of the positive-displacement type with a capacity and pressure adequate to produce the required atomization at the nozzle to obtain the ignition and combustion efficiencies specified herein. The fuel pump shall deliver fuel from the truck fuel tank to the heater at the maximum rate of fuel consumption.

3.7.12.5.2 Fuel pressure regulator. An adjustable fuel regulator shall maintain the required pressure to the burner nozzle. The fuel system shall be equipped with a device that will completely shut off fuel to the burner when the fuel pressure is insufficient for complete automization.

3.7.12.5.3 Fuel pressure gage. A pressure gage shall be provided of adequate range and size to permit determination and to allow adjustment of the operating fuel pressure. If practicable, the fuel pressure gage and regulator shall be mounted on the ground control panel of the spraying unit.

3.7.12.5.4 Fuel filter. A replaceable element-type fuel filter shall be provided for removing foreign particles from the fuel. The filter shall be provided with a sediment bowl that shall be fitted with a means for draining out the impurities that may accumulate during operation.

3.7.12.5.5 Heater fuel tank. The fuel tank of the truck shall be used as the fuel tank for the fluid heater.

3.7.12.6 Heater controls. All electric controls, other than the thermal switches, whose operation depends upon being mounted on the heater, shall be mounted on the ground control panel. A main heater switch shall be provided with a green indicator light to show when the heater is operating properly. A red trouble light shall be provided to indicate malfunctioning of the heater.

3.7.12.6.1 Temperature gages. Two dial-face-type temperature gages shall be mounted on the ground control panel to indicate the fluid temperature at the inlet and outlet of the heater. The inlet temperature shall be measured upstream from the heater bypass. The gages shall be capable of withstanding all temperatures encountered during storage and shall properly indicate all fluid temperatures encountered during the use of the spraying unit.

3.7.12.6.2 Heater outlet pressure gage. A dial-faced-type pressure gage similar to the main pump pressure gage shall be provided to indicate the fluid pressure at the outlet of the heater. The gage shall be mounted on the ground control panel close to the main pump pressure gage. The pressure drop through the heater, as indicated by these two gages, shall be used to indicate the necessity for periodically descaling the heater.

3.7.12.7 Heater electrical system. The electrical components of the fluid heater system shall operate on the 24-V electrical system of the auxiliary gasoline engine. The electrical components of the heater shall be radio shielded in accordance with the radio suppression requirements of the entire spraying unit.

3.7.12.8 Heater protection. The spraying unit shall be designed to provide protection against personnel accidentally burning themselves on heated surfaces as well as providing the necessary weatherproof protection to the heater components.

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3.7.12.9 Heater exhaust. The hot exhaust gages from the heater shall be directed up and to the outside of the vehicle with the duct so located as to eliminate the possibility of accidental burns to personnel during normal operation. The gas duct shall not be of flexible metal tubing. The exhaust outlet shall be provided with a means of preventing entrance of precipitation during storage and operation in the open under all weather conditions for extended periods of time. Components shall effectively resist rust and corrosion.

3.7.12.10 Heater dimensions and weight. The dimensions and weight of the heater assembly shall be the minimum consistent with the performance requirements specified herein.

3.7.12.11 Flow control switch. A flow control switch shall be installed in the system that will completely shut off the fuel to the burner and prevent burner ignition when liquid flow through the heater is stopped, such as by the sudden closing of the discharge valves and failure of the pressure relief valve to bypass fluid, or by pumping the tanks dry. Re-establishing flow of the fluid shall automatically establish the normal flow of fuel and restart the burner. If a magnetically-operated flow indicator is provided, a permanent magnet-type plug shall be incorporated into the plumbing system between the pump and flow control to remove iron filings caused by pump wear.

3.7.13 Air blast system. When specified (see 6.2), an air blast system shall be provided on the spraying unit. The air blast assembly shall be designed to deliver a high velocity airstream from a Type MA-1A gas turbine compressor conforming to MIL-C-4851 through a lightweight, maneuverable turret-type nozzle. The air blast assembly shall be designed for mounting on the lift and rotating structure of the spraying unit. All parts, fittings, controls, components, and accessories for mounting and safe operation shall be provided. Mounting provisions shall permit permanent installation. The Type MA-1A gas turbine compressor shall not be furnished with the air blast assembly. The mounted air blast assembly shall not limit nor interfere with the operational capabilities of the lift and rotating structure of the spraying unit.

3.7.13.1 Air discharge. The air blast assembly shall be designed to discharge a high-velocity airstream against aircraft lift and control surfaces. The air blast assembly shall be designed to accept the maximum flow, pressure, and temperature of the discharge air from the Type MA-1A compressor. The performance data of the Type MA-1A compressor is as specified in table II.

3.7.13.2 Discharge nozzle. A discharge nozzle shall be provided that will deliver a concentrated airstream at sonic velocity to efficiently remove accumulations of ice from aircraft lift and control surfaces at a minimum rate of 150 square feet per minute.

3.7.13.2.1 Nozzle extension. The nozzle shall extend to, but not beyond, the sides of the work platform when the nozzle is directed horizontally to either side.

3.7.13.2.2 Nozzle mounting. The nozzle shall be designed for mounting on and being controlled from the work platform of the spraying unit. Mounting shall allow complete freedom of movement of the nozzle through the area specified herein at any position of the platform and boom. The area covered by the nozzle shall be horizontally from side-to-side through 180°, and from a minimum of 60° above horizontal to the front down and to the rear through an arc of approximately 210°.

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TABLE II. Compressor performance data.

	Ambient Air Temperature		
	+59°F	+130°F	-65°F
Airflow[1]	119 pm	90 pm	170 pm
Air pressure[1]	50 psia	45 psia (min)	60 psia (min)
Air temperature[1]	+370°F	+500°F (max)	+225°F (min)

[1]Measured at the end of a 30-foot hose supplied with the Type MA-1A compressor.

3.7.13.2.3 Nozzle control. The operator's nozzle control shall effectively control and direct the nozzle assembly under all conditions of operation. The nozzle shall remain in the last selected position when the control is released. The control system shall prevent rapid, uncontrolled movement of the nozzle. If required to assure complete ease of operation, the controls shall be provided with a power-assist mechanism. The nozzle control shall be designed for mounting on the operator's platform convenient to the lift and rotation controls.

3.7.13.3 Remote control. A remote control shall be provided to start and stop, and load and unload the gas turbine compressor. Controls shall include a master switch and a load light. The wiring circuit and part numbers shall be as shown on sketch A of figure 3. The remote control shall be designed for mounting on the operator's platform convenient to the nozzle controls. The remote control cable shall connect the platform controls to the gas turbine compressor through the connector provided for this purpose. One set of adapters as shown in sketches B and C of figure 3 shall be provided with each remote control and shall be permanently marked for identification by reference to the drawing number (see 6.3).

3.7.13.4 Fuel tank. An 80-gallon aluminum fuel tank shall be mounted on the opposite side of the turntable from the gas turbine compressor. The fuel tank cap shall be attached to the fuel tank by a metal chain. The fuel tank shall meet all other requirements specified in 3.7.1.2.

3.7.13.4.1 Fuel filter and boost pump. A fuel filter and boost pump, identical to and interchangeable with those required on the gas turbine compressor, shall be provided.

3.7.13.4.2 Fuel line. A fuel line shall be provided to deliver fuel from the fuel tank to the gas turbine compressor. The fuel line shall be capable of supplying fuel to the turbine at its maximum rate of consumption. A quick connector shall be provided on the end of the fuel line to mate with and connect to the turbine fuel connector. The length of the fuel line shall permit easy connecting without the use of extensions or adapters.

3.7.13.5 Battery connector. A 28-V, d-c, electrical power cable assembly conforming to MS90347 shall be provided. The cable assembly shall be of sufficient length to provide for easy connection of the auxiliary batteries to the mounted gas turbine compressor. Operating instructions shall state that the power cable shall be disconnected from the compressor after it has

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been started and before the articulating booms of the spray outfit are elevated or rotated. These instructions shall be prepared on permanently etched or engraved metal or plastic placards and securely attached to the quick-connect end of the cable.

3.7.13.6 Mounting provisions. Mounting of the tubing, couplings, controls, nozzle, et cetera, shall not interfere with the normal operation of the structure. All elements of the assembly shall be attached by threaded connections. Dust caps shall be attached to the vehicle for the air, fuel, and electrical leads to retain the component in a position for quick connecting to its part.

3.7.13.6.1 Compressor mounting provisions. The gas turbine compressor mounting frame and supports shall be designed for a normal load of 700 pounds. Compressor dimensions and location of mounting provisions shall be as specified in MIL-C-4851. The mounting with its frame and support shall be furnished and shall mount on the turntable without interfering with its continuous rotation. The fuel line fitting and remote control connector shall mate with the corresponding part on the gas turbine compressor and shall be of such length and location as to provide for easy connecting.

3.7.13.6.1.1 Compressor swivel elbow. A 90° swivel elbow shall replace the flange fitting conforming to AiResearch part No. 101 (see 3.7.13.8). An adjustable support shall be provided to sustain the weight of the swivel elbow and adjacent tubing under all operating conditions.

3.7.13.7 Tubing. Lightweight, metallic, 3-1/2-inch OD tubing shall be used where flexing is not encountered. Flexible pneumatic duct conforming to MIL-D-26124 shall be used between the two booms. The inside diameter of the tubing shall be the maximum consistent with the requirements herein. A V-band clamp assembly shall be furnished to connect to the gas turbine compressor. A telescoping tube shall be provided on the lower boom.

3.7.13.7.1 Lateral fittings. An aluminum lateral fitting shall be incorporated in the vertical section at the rear of the platform. The lateral shall be provided with a valve to direct the air straight through the fitting to the air nozzle or out the lateral for engine starting. A lock shall be provided to retain the valve in the selected position. The lateral shall be provided with a marman flange and a cap to form a leakproof connection.

3.7.13.8 Swivel elbows. Swivel elbows shall be used where rotation or hinge action is encountered. The swivel elbows shall be leakproof, durable, and shall require a minimum of torque under operating conditions. A swivel elbow shall be provided between the turbine and the lower boom and between the upper boom and the work platform.

3.7.13.9 Lifting strap. A lifting strap having a capacity of at least 700 pounds shall be provided. The lifting strap shall fasten to the work platform and permit lifting the gas turbine compressor onto its mounting brackets.

3.7.13.9.1 The lower end of the lifting strap shall have three substraps. Two substraps shall connect to the gas turbine compressor lifting connectors and the third substrap shall be used to guide the gas turbine compressor into position. The length of the third substrap shall permit guiding personnel to remain at a safe distance while preventing the gas turbine compressor from swinging into the elevating booms.

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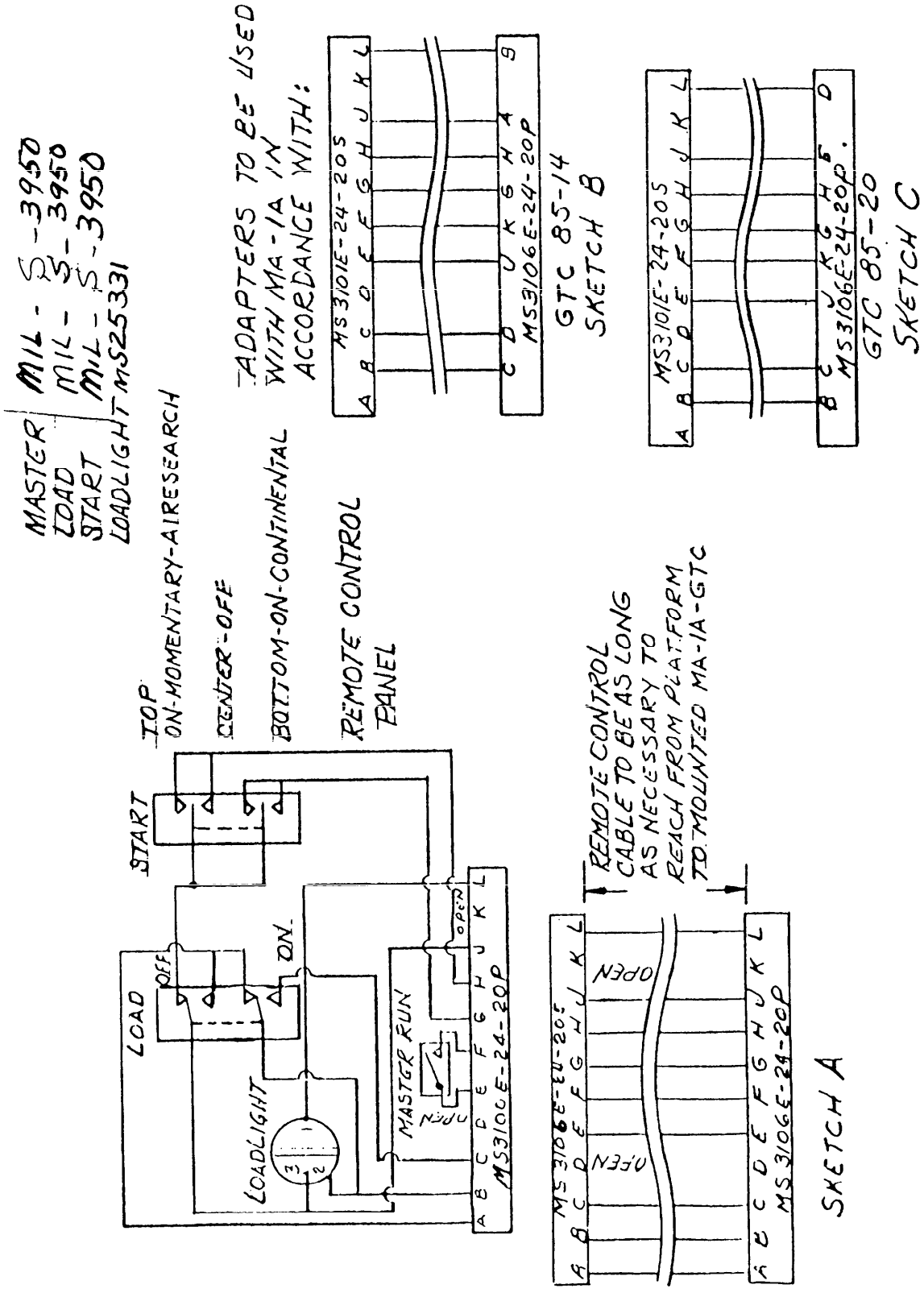


Figure 3. Wiring Diagram for Remote Control Panel and Cable

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3.7.13.10 Tube-to-duct connectors. V-band clamp assemblies and duct connectors conforming to MIL-D-26124 shall be provided to securely clamp the pneumatic duct to the metal tubing.

3.7.13.11 Accessories. With the exception of the gas turbine compressor, all connections, fittings, and accessories required to mount and operate the air blast assembly shall be provided.

3.7.13.12 Storage provisions. A means shall be provided for storing and protecting the connecting tubing (swivel elbow), fuel line, and electrical connectors when not in use. Provisions shall be made to remove and safely store the nozzle control assembly and the platform tubing section on the vehicle.

3.7.13.13 Access ladder. A metal ladder shall be provided on the rear of the equipment for personnel to climb safely from ground level to the equipment flooring. The steps or rungs shall have nonskid surfaces. The ladder shall be located at the right rear corner of the vehicle and may be hinged to permit raising to a vertical position when not in use.

3.7.13.14 Folding platform. A folding platform shall be provided and shall be hinged to the equipment flooring. The platform shall overhang the right side of the vehicle and extend along the side from the right rear corner. The platform shall be of sufficient area to permit one person to move freely while installing and removing the compressor and for other required functions. The minimim dimensions of the working area shall be 24 by 72 inches. The platform shall be designed for a normal working load of 500 pounds. With the exception of the dimensions, the platform shall meet the requirements specified in 3.7.4.2. When raised to a vertical position, it shall retract to within the overall dimensions of the spraying unit. Provisions shall be made to lock the platform in a vertical position. Neither the platform nor the ladder shall limit nor interfere with the operational capabilities of the lift and rotating structure of the spraying unit.

3.7.13.14.1 Precautionary placard. The following placard, permanently and legibly filled in, shall be securely attached to the front, back, and outer side of the platform:

WARNING

RATED PLATFORM LOAD - 500 POUNDS
PLATFORM AND LADDER[1] MUST BE LOCKED
IN VERTICAL POSITION DURING OPERATION
OF THE SPRAYING UNIT.

[1]IF LADDER IS HINGED.

3.8 Handbook compartment. A compartment shall be provided in the truck cab for containing handbooks covering the operation, service, and repair of the spraying unit. The compartment shall be a minimum size of 9 by 11-1/2 by 2 inches.

3.9 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern manufacturer's part numbers and changes thereto.

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3.10 Electromagnetic interference. The spraying unit shall comply with the requirements specified in MIL-STD-826.

3.11 Noise level. The noise level of the spraying unit, excluding the air blast system, shall not exceed the limits specified herein for various frequency bands under any operating conditions specified herein when measured approximately 5 feet above ground level at a distance of not more than 25 feet in any direction from the spraying unit or at the ground operator's position. Maximum allowable noise level limits shall be as follows:

Frequency Band (Cycles Per Second)	Noise Level Limit (DB Above 0.0002 Dynes/CM^2)
20 to 75	108
75 to 150	98
150 to 300	90
300 to 600	83
600 to 1,200	78
1,200 to 2,400	75
2,400 to 4,800	72
4,800 to 10,000	71.

3.12 Dimensions. The overall dimensions of the spraying unit, including the truck and with the booms and stabilizers in the travelling position, shall not exceed a width of 96 inches and a length and height consistent with the air transportability requirements specified in 3.5.12.

3.13 Threads. All screw and pipe threads shall conform to FED-STD-H28.

3.14 Finishes and protective coating.

3.14.1 Cleaning, painting, plating, and anodic coatings. Cleaning, painting, plating, and anodic coatings shall be in accordance with MIL-STD-808. All surfaces of the spraying unit shall be finished in yellow enamel, color code DG.

3.14.1.1 High-temperature-resistant finishes. Heater exhaust ducts and other surfaces subjected to high temperatures shall be finished in accordance with procedure FF-903 of MIL-STD-808.

3.14.1.2 Truck chassis. The truck chassis shall be painted yellow, color code DG, conforming to MIL-STD-808. Exterior finish painting of the frame, springs, transmission, propeller shaft, and axles (not wheels) may be black, except at a location on the truck chassis where this would not be appropriate in that it would detract objectionably from the appearance of the yellow exterior finish. When appropriate, such items as running boards, taillamps, and outside mirrors may be black. Plated, polished finishes need not be painted. Interior finish of the cab may be the manufacturer's standard, preferably a color that blends with the yellow exterior finish. Sheet-metal surfaces not exposed for finish painting and subject to rusting, such as cab door panel interior surfaces, shall be properly treated or coated in accordance with best commercial practices to effectively prevent rusting.

3.14.2 Stencils and marking. Stenciled markings shall be in accordance with MIL-STD-808 and shall be conspicuously located near the item concerned.

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Control handles and other parts shall be painted black as appropriate to attract attention, provide a neat appearance, or cover surfaces that are subject to having their finish worn off during use.

3.15 Markings.

3.15.1 Truck. The following markings shall be applied to each vehicle: The agency identification and the registration number shall be centered on each front door and on the rear of the vehicle in two separate lines. These markings shall be affixed to the vehicle by means of reflectorized tape in accordance with MIL-STD-808. The marking U.S. AIR FORCE shall appear on the first line in 4-inch letters. The registration number (to be furnished by the procuring activity) shall appear on the second line in 3-inch letters (see 6.2). AF/S32M-1 shall appear on the third line in letters 2 inches high. There shall be a 1-inch space between lines. The correct operating tire pressure shall be marked on the left front pillar post in letters 1/2 inch high. The gross weight of the vehicle, lettered 1-inch high, shall appear low on the tank cover. The NATO symbol (a black, color code PG in accordance with MIL-STD-808, triangle that is 6 inches per side and that points up) shall be applied in a conspicuous location high on each side and on the rear of the equipment. Immediately below the symbol in legible-size letters shall appear The words CLEANING - DEICING - DECONTAMINATING on the first line, and NATO CODE. No. 742 (DEICING) on the second line. The loction of the center of gravity shall be marked on both sides of the vehicle.

3.15.1.1 Truck data plates. The manufacturer's standard nameplates and data plates shall be acceptable and shall be permanently attached to the truck. The data plate shall show the contract number, date of delivery, and inspector's stamp.

3.15.2 Spraying unit. Placards and markings shall be applied to the spraying unit as follows:

3.15.2.1 Operating instructions. Instructions for step-by-step operation of each system of the spraying unit, followed by instructions for adjustments, drainage, and inspections shall be prepared in an abbreviated form on permanently etched or engraved metal or plastic instruction placards. The placards shall be permanently mounted on the control panel.

3.15.2.2 Precautionary placard. The following precautionary placard, permanently and legibly filled in with the required information, shall be securely attached to the ground control panel and on the platform, preferably adjacent to the operating controls.

WARNING

RATED PLATFORM LOAD, STABILIZER JACKS DOWN, 750 POUNDS

RATED PLATFORM LOAD, MOBILE OPERATION WITH SPRING LOCKNUTS ENGAGED, 500 POUNDS

USE CAUTION IN OPERATING PLATFORM WITH STABILIZER JACKS UP DURING MOBILE OPERATION OF THE VEHICLE.

3.15.2.3 Other markings. Other markings containing operating instructions or precautions, drain locations, et cetera, essential to safe operation of the

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spraying unit shall be conspicuously located and permanently attached, preferably adjacent to the item concerned.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The spraying unit, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

3.17.1 Riveting. Riveting operations shall be performed to insure that the rivets are tight and satisfactorily headed.

3.17.2 Welding. Where welding, brazing, or soldering is used in the fabrication of the spraying unit, the process used shall be in accordance with the highest commercial standards and in accordance with the latest applicable codes of the American Welding Society. No welding shall be resorted to as a repair measure, except that small blowholes in castings may be filled when not in critical areas, provided that the responsible inspector is fully informed of all pertinent details and approves the repair procedure prior to its accomplishment. All welded joints shall be inspected to insure freedom from cracks, oxide inclusions, or injurious porosity.

3.17.3 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the spraying unit, they shall be held or limited accordingly.

3.17.4 Screw assemblies. Assembly screws and bolts shall be tight.

3.17.5 Cleaning. The spraying unit shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the spraying unit shall be classified as follows:

a. Preproduction testing (see 4.4).

b. Acceptance tests (see 4.5).

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4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Unless otherwise specified, the tests shall be conducted at standard sea level atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 70° F).

4.3.2 Test time. Test periods shorter than 1 hour, including intervals of the endurance test time less than 1 hour terminated by a parts failure, shall not be credited as endurance test time. Endurance test time shall be credited in 1/2-hour intervals.

4.3.3 Apparatus and instruments. Apparatus and instruments used in conjunction with the testing specified herein shall be of laboratory precision type and shall be checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation. They shall be calibrated at properly spaced intervals to insure laboratory accuracy.

4.3.4. Fuel consumption. Fuel consumption measurements shall be made by either the volume or weight method. The quantity selected for the volume or weight method shall be such that each reading will cover an elapsed time of at least 15 minutes. Flowmeter readings for fuel consumption shall be used only when the flowmeter and use thereof has been specially approved by the procuring activity. All expressions of quantity shall be recorded on a weight basis.

4.3.5 Liquid flow rates. Liquid flow rates shall be determined by weighing the liquid flowing over a period of not less than 1 minute or measuring the time to fill or empty a container of known or calibrated volume. Unless the flowmeter has a certified accuracy of 3 percent or better and use thereof has been specifically approved by the procuring activity, flowmeter readings shall not be used.

4.3.6 Temperatures. Temperatures shall be measured by means of appropriately located thermocouples or thermometers. Insofar as is practical, thermocouples shall be isolated from contact with other metals. Temperatures shall be expressed in degrees Fahrenheit.

4.3.7 Pressures. Insofar as is practicable, pressures and pressure differentials shall be measured by means of liquid manometers. Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. Gages shall be of laboratory-type having a dial diameter of not less than 4 inches.

4.3.7.1 Discharge pressures. Discharge pressures specified herein are those measured at the spray guns.

4.3.8 Preparation for test. Prior to testing, the spraying unit shall be serviced with fuels and lubricants in accordance with the instructions of the manufacturer.

4.3.9 Preliminary run-in. Any preliminary run-in of the spraying unit as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the spraying unit for tests.

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4.3.10 Data. During the testing specified herein, sufficient data shall be recorded at proper intervals to insure compliance of the spraying unit with the performance requirements specified. Data shall be recorded every hour during the endurance testing. Recorded data shall include the date, time, and ambient temperature at which the test is conducted as well as the data needed to accurately calculate the performance rates, efficiencies, et cetera, to be determined by specific test.

4.3.11 Inspection, servicing, and adjustments. Planned stops shall be made at intervals during the endurance tests for inspection, servicing, and adjustments. Records shall be kept of all adjustments made regardless of how minor the adjustment. Major adjustments shall be cause for rejection of the spraying unit.

4.3.12 When a cleaning compound system is specified, the tests specified in 4.6.2.6, 4.6.14, and 4.6.17.3 b(2) shall be conducted.

4.3.13 When an air blast system is specified, the tests specified in 4.6.2.7, 4.6.2.8, 4.6.16.1.1, and 4.6.17.2 shall be conducted.

4.4 Preproduction testing.

4.4.1 Preproduction test sample. The preproduction test sample shall consist of a model representative of the production equipment. It shall be tested at a site selected by the contractor under supervision of the procuring activity (see 6.2).

4.4.2 Preproduction test report. The contractor shall prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.4.2.1 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements of this specification.

a. Description of failed part(s) and cause of failure.

b. Corrective action taken and a prediction of how this action will affect reliability and maintenance.

c. The number of man-hours individually expended performing servicing, adjustments, and maintenance during the testing period.

d. Comments shall include remarks made by test personnel pertaining to the unduly difficult, time-consuming, and frequency of servicings, adjustments and maintenance of the unit.

e. Recommendations shall include comments on features that if modified would improve performance, reliability, or maintainability.

f. Comments on operation or operating procedures that should be utilized or avoided in field use that will improve reliability.

4.4.2.2 Maintainability verification. Verification of the maintainability requirements shall be in accordance with MIL-STD-471.

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4.4.3 Preproduction tests. The preproduction tests shall consist of all test specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests.

b. Sampling plan and tests.

4.5.1 Individual tests. Each spraying unit shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1).

b. Performance demonstration (see 4.6.2).

c. Boom assembly (see 4.6.2.1).

d. Main spraying system (see 4.6.2.2).

e. Hot water spraying (see 4.6.2.3).

f. Compound system (see 4.6.2.6).

g. Air blast system (see 4.6.2.8).

4.5.2 Sampling plan and tests.

4.5.2.1 Sampling plan. One spraying unit shall be selected at random from every 50 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Dependability (see 4.6.3).

b. Creep test (see 4.6.4).

c. Mobility tests (a minimum of 5 percent of the distances specified for the tests in 4.6.5.1) (see 4.6.5).

4.6 Test methods.

4.6.1 Examination of product. The spraying unit shall be inspected to determine compliance with applicable drawings and the requirements specified herein with respect to parts, materials, design, dimensions, weights, finishes, markings, and the functional operation of all components, assemblies, accessories, and controls.

4.6.2 Performance demonstration. The spraying unit shall be operated under the following conditions to insure that the performances of the various components comply with the requirements herein. During this performance demonstration, the auxiliary engine and other components shall be operated according to the instructions provided on the spraying unit.

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4.6.2.1 Boom assembly. The boom assembly shall be operated from the platform and from the ground control panel with a rated load of 750 pounds on the platform to insure compliance with the requirements of 3.7.3.1, 3.7.3.2, and 3.7.6.3. The loaded platform shall be simultaneously raised from the lowest to the highest position and rotated through 360° in not less than 45 seconds nor more than 60 seconds.

4.6.2.2 Main spraying system. The main pumping system shall deliver cold water through the high-pressure hoses, spray guns, and nozzles at the volumes, pressures, and spray patterns specified in 3.7.10.3.1 and 3.7.10.3.2. While delivery through the spray guns is being measured, the main pump shall deliver at least 50 gpm at nozzle discharge pressures of 100 and 150 ± 5 psig.

4.6.2.3 Hot water spraying. Hot water at a temperature of 180° to 200° F shall be delivered through each of the high-pressure delivery hoses and spray guns at a flow rate of at least 10 gpm and a discharge pressure of 100 ± 5 psig. Delivery shall be made through all combinations of one to three spray guns during this test.

4.6.2.4 Heater efficiency. The thermal efficiency of the fluid heater shall be determined. Water temperature at the inlet and outlet of the heater shall be measured to determine the temperature rise. The rate of water flow through the heater shall be measured. Using the temperature rise and the flow rate, the BTU output shall be determined. The output capacity of the heater shall be a minimum of 1,000,000 BTU per hour. The BTU input shall be determined using the measured rate of fuel consumption and an acceptable heat content value for the fuel used. The minimum overall thermal efficiency shall be not less than 75 percent.

4.6.2.5 High-pressure cleaning. A high-pressure hot water rinse shall be delivered through each of the three high-pressure hoses and spray guns at a minimum rate of 10 gpm and a discharge pressure of 150 ± 5 psig. Delivery shall be made through all combinations of one to three spray guns. The rinse shall be heated to 180° to 200° F.

4.6.2.6 Cleaning compound spraying system. The compound spraying system shall deliver a combined flow of at least 10 gpm from the two hoses and spray guns at a discharge pressure of 50 ± 5 psig. Each of the spray guns and nozzles shall deliver at the flow, pressure, and spray pattern specified in 3.7.11.4.

4.6.2.7 Normal operation. With the air blast assembly mounted on the elevating structure and work platform of a de-icing unit and with a mounted and connected Type MA-1A gas turbine compressor conforming to MIL-C-4851, the assembly shall be checked for ease and accuracy of nozzle control and adequacy of delivery of high-velocity air while the gas turbine compressor is operating at maximum rated capacity.

4.6.2.8 Nozzle direction control. With the gas turbine compressor delivering air at rated capacity, the nozzle control mechanism used to direct the nozzle shall be thoroughly tested for ease of operation and dependability. There shall be no change in nozzle elevation or azimuth when the control is released.

4.6.3 Dependability. The spraying unit shall be subjected to 10 percent of the endurance tests specified under 4.6.17 to determine the dependability of the unit. Failure, deterioration, or undue wear resulting from this test

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shall be cause for rejection. A reduction in performance greater than 1 percent shall also be cause for rejection. Immediately following completion of 10 percent of the boom assembly endurance test, the spraying unit shall be subjected to the test specified in 4.6.4.

4.6.4 Creep test. With a 750-pound load on the platform, the booms shall be rotated 90° and elevated to form an angle 45° with the horizontal plane. The front wheels of the truck shall be raised to incline the turntable at an angle of at least 3° with the horizontal plane. The unactuated horizontal movement and vertical creep of the platform shall not exceed 2 inches when measured over 1 hour.

4.6.5 Mobility tests.

4.6.5.1 Road. The spraying unit, with all tanks filled, shall be driven over level, paved highways at 50 mph for a distance of 50 miles. At least 100 short radius right turns, 100 short radius left turns, and 10 sudden stops from 20 mph shall be accomplished during this test. The fluid tanks shall be drained and the spraying unit shall then exhibit its capability to creep up a 20° slope in both forward and reverse gears. Excessive wear or other damage to the truck running gear or damage to or instability of the mounted equipment during the road tests shall be cause for rejection. Upon completion of the road tests, the spraying unit shall be subjected to the performance tests specified in 4.6.2.1, 4.6.2.5, and 4.6.2.6.

4.6.5.2 Brakes. The service brakes of the truck shall stop the fully-loaded spraying unit within a distance of 25 feet when traveling at a speed of 20 mph on dry, level concrete. It shall be demonstrated that the service brakes and parking brake will hold the spraying unit in an air-transportable weight condition on a 20° or 36-percent grade.

4.6.5.3 Mobile operation. The spraying unit shall be operated to determine compliance with 3.5.11.2. The boom assembly shall be operated throughout its range while the truck is being driven under 5 mph as if moving around a large aircraft. Any evidence of instability or overturning of the spraying unit during this test shall be cause for rejection.

4.6.6 Air transportability. The spraying unit shall be subjected to the ramp test and loading demonstration specified in MIL-A-8421. Computations shall be acceptable from a design standpoint and may be substituted for the acceleration and load tests specified in MIL-A-8421.

4.6.7 Initial inspection. Each spraying unit submitted for tests shall be disassembled to the maximum practical extent and completely inspected. Each component shall be checked as to weight, measurements, et cetera, to determine compliance with the requirements specified herein and to correctly determine wear or other conditions that may be the result of the preproduction tests. The spraying unit shall be inspected to determine compliance to the applicable drawings. Following this inspection, the spraying unit shall be reassembled and subjected to the remainder of the preproduction tests.

4.6.8 Electromagnetic interference. The spraying unit shall be tested for radiated interference in accordance with MIL-STD-826 for class Ib equipment. The antenna shall be positioned 6 feet from the outer perimeter of the unit at the point of maximum radiated interference.

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4.6.9 Noise level test. The noise level of the spraying unit shall be measured at a number of points on a perimeter not more than 25 feet from the spraying unit and at the ground operator's position. Measurements shall be taken out-of-doors at a distance of approximately 5 feet above the ground. A Type 759A General Radio Company sound level meter, or equal, shall be used to test for noise level. The noise level of the spraying unit (without the air blast system in operation) shall not exceed the limits specified in 3.11 under any operating condition.

4.6.10 Operation in the tilted position. The complete spraying unit shall be tilted in at least four different directions, 90° apart, on a 3° incline from the true horizontal plane. The spraying unit shall be started and operated for 30 minutes in each position. The spraying unit shall perform satisfactorily in every respect as specified in 4.6.2. Tests shall be conducted to determine that complete lubrication is maintained during all phases of the tilted operation.

4.6.11 Platform overload. With the fluid tanks empty and the stabilizer jacks extended, the components of the boom assembly shall be tested for overload conditions. A 1,500-pound evenly distributed load shall be placed on the work platform at various positions throughout the range of movement of the platform, including the position most conducive to tipping the spraying unit. The hydraulic pressure of the spraying unit may be increased to move the 1,500-pound load to the various positions required in this test. There shall be no evidence of upsetting or instability of the spraying unit and no component of the boom assembly shall show evidence of permanent deformation after completion of the test.

4.6.12 Platform flooring load. The expanded metal flooring of the work platform shall be subjected to a vertical load of 300 pounds distributed over an area of 25 square inches. No point on the flooring shall deflect more than 1/4 inch, and this loading shall not cause permanent deformation of the flooring.

4.6.13 Platform guardrails and safety belt load. The platform guardrails shall be subjected to a load of 500 pounds at various positions and in horizontal and vertically downward directions. There shall be no permanent deformation of any member of the guardrails or the platform structure as a result of this test. Each safety belt shall support a load of 500 pounds without failure of permanent deformation of any part which would render the belt or its fastenings inoperable.

4.6.14 Hydrostatic pressure. All components of the fluid delivery system shall be subjected to hydrostatic pressures of twice the normal working pressure of each system. There shall be no evidence of leakage or failure of any component as a result of this test.

4.6.15 Fluid heater. The fluid heater assembly shall be subjected to the following tests:

4.6.15.1 Explosion. With the combustion chamber and heat exchanger components at room temperature, a gaseous vapor, in a ratio of 0.085 to 0.095 of fuel to air, shall be introduced into the combustion chamber. The chamber shall be purged with this mixture to the extent of at least 10 times the volume of the

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chamber. The mixture shall be ignited with the heater ignition system. This procedure shall be repeated 50 times. The heater shall show no deleterious effects of the explosions and thereafter shall complete all ensuring tests in a satisfactory manner.

4.6.15.2 Static pressure. The pressure side of the heater system shall be subjected to a hydrostatic test of twice the maximum working pressure as specified under 4.6.14. The heater coil shall be subjected to a static pressure of 1,000 psig as specified in 3.7.12.1.

4.6.15.3 Safety controls. The burner safety controls shall prevent any burner flash or explosion during the following tests while burning gasoline. The safety controls shall also automatically re-establish the burner flame within 1 minute after return to normal operation of the heater. After the heater has been in operation for a minimum of 1 hour to ensure normal combustion chamber temperatures, malfunction of the heater shall be caused by the following methods to determine proper function of the safety controls:

4.6.15.3.1 Fuel system interruption. While the heater is in full operation, the flow of fuel to the burner shall be interrupted for 10 seconds and flow re-established. After the fuel system has been interrupted, the safety controls shall automatically permit a complete purging of the combustion chamber before the ignition of the fuel can be regained.

4.6.15.3.2 Water shortage or failure. While the heater is in full operation, the water supply shall be turned off until the flow-control switch stops the burner. Immediately after the burner has been stopped, the water supply to the heater shall be turned on. The heater shall automatically regain safe burner operation.

4.6.15.3.3 Shortage or depletion of fuel supply. While the heater is in full operation, the fuel supply to the heater fuel pump shall be depleted by opening a line or emptying the fuel tank causing the low fuel pressure cutoff device to operate. The heater light shall indicate loss of burner operation. When the fuel supply is replenished, the burner shall start and operate satisfactorily.

4.6.15.3.4 Ignition failure due to faulty components. The ignition electrode shall be grounded to simulate a cracked electrode or burned-out transformer or vibrator. The safety controls shall prevent fuel from flowing to the burner under these conditions and the signal light shall indicate burner failure.

4.6.16 Environmental. The spraying unit shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810. Upon completion of each test, the spraying unit shall be subjected to the tests specified in 4.6.2. Failure of the spraying unit to operate as required during or after each exposure, failure of any item or assembly due to deterioration from such exposure, or formation of any undesirable condition that cannot be remedied by routine maintenance procedures shall be cause for rejection.

4.6.16.1 Low temperature. The spraying unit shall be subjected to the test specified in procedure I. The spray fluid systems shall be filled with anti-freeze solutions (glycol-water solutions) protected to prevent freezing of the fluid in the equipment. The hydraulic system of the spraying unit shall

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be filled with a mixture of 75 percent hydraulic fluid conforming to MIL-H-5606 and 25 percent aliphatic naphtha conforming to TT-N-95. Prior to starting the test, the spraying unit shall be cold-soaked at -65° F, with winterization systems in operation, until all temperatures are stabilized. The truck and other components of the spraying unit shall start within 15 minutes. The load on the auxiliary engine shall be slowly increased to rated output so that the spraying unit will perform as specified in 4.6.2 within 15 minutes after starting at an ambient temperature of -65° F.

4.6.16.1.1 Ice removal. Ice shall be formed on the empennage and wing of a large cargo or bomber aircraft or equivalent test surfaces. Ice formed shall be 1/4, 1/2, and 1 inch thick. The ability of the air blast assembly to be maneuvered and to remove ice as specified in 3.7.13.2 shall be checked. Ice shall be formed and removed at 20° ± 5° F.

4.6.16.2 High temperature. The spraying unit shall be subjected to the test specified in procedure I. Upon completion of the exposure to high temperature, the ambient temperature shall be lowered to 130° F. The spraying unit shall then be tested as specified in 4.6.2. There shall be no evidence of softening of thermoplastic materials or other damage due to the high temperatures.

4.6.16.3 Humidity. Components of the spraying unit which are subject to deterioration as a result of high humidity shall be subjected to humidity in accordance with the procedures specified.

4.6.16.3.1 At the discretion of the procuring activity, the humidity test may be required only for the critical or operating components subject to deterioration by the elements.

4.6.16.4 Altitude. The spraying unit shall be operated continuously for 5 hours at a pressure altitude of 6,000 feet, and shall deliver rated output through each delivery system as specified in 4.6.2. The ambient temperature shall not fall below 48° F during the tests. The gasoline engine shall show no signs of overload or overheating during the tests. If an altitude chamber is not available, an evaluation of altitude characteristics supported by detailed performance data of the engine and pumps shall be submitted.

4.6.17 Endurance tests. The spraying unit shall be subjected to the following endurance tests. During these tests, the ambient air temperature shall be maintained at 60° F, or higher.

4.6.17.1 Boom assembly endurance. The boom assembly shall be subjected to the following operating cycles and shall be capable of normal operation after 500 complete cycles. During each cycle, the turntable shall be rotated 360° and the upper boom shall be operated to move the platform from the lowest to the highest and back to the lowest platform positions. After every 10 cycles, the outriggers shall be retracted and extended, the rear spring lockouts shall be disengaged and engaged, and the booms shall be folded to the traveling position and raised to lower the platform to the lowest position. During this test, one-half of the boom cycles shall be controlled from the platform and one-half from the ground control panel. One-half of the turntable rotation shall be in each direction. For 250 cycles, there shall be a 500-pound load on the platform and for the other 250 cycles there shall be a 750-pound load on the platform. The boom assembly shall be operated continuously for a minimum of 50 cycles to be credited to this test.

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4.6.17.1.1 Immediately after the test specified in 4.6.17.1, the spraying unit shall be subjected to the test specified in 4.6.4.

4.6.17.2 Air blast nozzle control assembly. While the gas turbine compressor is operating at rated capacity, the air blast nozzle control assembly shall be subjected to the following cycle of tests and shall be capable of normal operation after 100 complete cycles:

a. Fifty cycles - scribing a vertical figure eight

b. Fifty cycles - scribing a horizontal figure eight

4.6.17.3 Spraying systems endurance. The spraying systems shall be subjected to the following 200-hour endurance test. Restrictors shall be used in the delivery systems to establish the specified test pressures. Test data shall be recorded every hour during this test. The endurance test for the spraying systems shall be conducted for the periods of time and under the operating conditions specified.

a. One hundred and fifty hours delivering water heated to between 180° and 200° F at a discharge pressure of 100 ± 5 psig through the high-pressure hoses, spray guns, and nozzles. Delivery shall be made through all three hoses at a combined rate of at least 30 gpm for 75 hours and through each hose separately for 25 hours at a flow rate of at least 10 gpm. During at least 50 hours of this test, the main pump shall draw water from an auxiliary source through the 15-foot length of suction hose at a suction head of 15 feet of water.

b. (1) Fifty hours delivering high-pressure water through the high-pressure hoses, spray guns, and nozzles at a discharge pressure of 150 ± 5 psig. Delivery shall be made through all three hoses at a combined flow rate of at least 30 gpm for 20 hours and through each hose separately for 10 hours at a flow rate of at least 10 gpm. The water shall be heated to 180° to 200° F. (2) During the same 50 hours, the spraying unit shall deliver cleaning solution at a discharge pressure of 50 ± 5 psig. Compound delivery shall be made through the two hoses at a combined rate of at least 10 gpm for 30 hours and through each hose separately for 10 hours at a flow rate of at least 5 gpm.

4.6.17.4 Leakage inspection. During the endurance tests, the spraying unit shall be inspected for leakage. Fluid lines, fittings, pumps, and other components shall show no leakage. However, if pumps and other equipment have packed glands that can be tightened during the endurance test to stop leakage that may have occurred, such adjustments will be acceptable.

4.6.17.5 Performance check. Immediately following the endurance tests and without intervening reconditioning, the spraying unit shall be subjected to the performance tests specified in 4.6.2. This data shall be compared with the data originally recorded under 4.6.2. Failure, deterioration, or undue wear resulting from the endurance test shall be cause for rejection. A reduction in performance greater than 5 percent below specified requirements shall also be cause for rejection.

4.6.18 Windload. The rear spring lockouts shall be engaged and the stabilizer jacks extended to provide maximum stability. With a 750-pound load on the platform, the platform shall be extended to maximum height. The turntable shall be rotated 90°, or to that position that will present the greatest boom

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area to a wind coming from either side of the vehicle. The unit shall then be subjected to the windload specified in 3.6. Test winds may be generated by artificial means or by applying equivalent simulated loads. After removal of these loads, there shall be no evidence of damage or permanent deformation of any part, section, or member of the spraying unit (see 6.2).

4.6.19 Serviceability. The spraying unit shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of decontamination residue, dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.6.20 Final inspection. Upon completion of the tests, the spraying unit shall be disassembled to the maximum practical extent for a thorough examination of all parts to disclose failure, distortion, excessive wear, and other unsatisfactory conditions. When measurements are made to determine wear, they shall be compared with similar measurements made during the initial inspection. All parts shall be in an acceptable condition. Wear or distortion that requires rework or replacement of the parts affected shall be cause for considering the part affected as having failed to complete the test satisfactorily. Spraying units not conforming to the requirements herein shall be rejection.

4.7 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.7.1 Individual tests may continue. For operation reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.7.2 Defects in items already accepted. The investigation of test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. The spraying unit shall be preserved and packaged in accordance with MIL-STD-162, except as follows:

5.1.1.1 Chassis. The truck chassis shall be preserved and packaged in accordance with MIL-STD-281 level A (mobile).

5.1.1.2 Hydraulic system of the boom assembly. Preservatives shall not be applied to interior surfaces of the hydraulic system. The hydraulic fluid supply tank shall be filled to the proper operating level with hydraulic fluid required by the technical manual supplied with the equipment. Pistons shall be fully retracted into cylinders, or as far as the folded storage

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appropriate soil group classification is then assigned to the soil material- the

position of the boom assembly permits, and wired or strapped into the retracted position. Wooden blocks shall be secured in place by stapling the wire or strapping thereto. Exposed surfaces of piston rods shall be coated with Type P-6 preservative; spirally wrapped with type I, grade A, class 1, greaseproof barrier-material; overwrapped with type I, grade C, class 1, greaseproof barrier-material secured in place with cord; and sealed with dipcoating sealing compound. All assemblies actuated by the hydraulic system shall be wired or strapped to the main unit to prevent movement. Hydraulic operating valve controls shall be secured against movement and exposed portions of the controls shall be coated with Type P-6 preservative; spirally wrapped with type I, grade A, class 1, greaseproof barrier-material; overwrapped with type I, grade C, class 1, greaseproof barrier-material secured in place with cord; and sealed with dipcoating sealing compound. Plugs, hose, and similar parts of the hydraulic system shall not be detached unless the parts that the system actuated must be detached to reduce cubage. All openings, ends of hose, and connections, where removals are made, shall be sealed with type I, grade B, pressure-sensitive tape conforming to PPP-T-60.

5.1.1.3 Delivery hoses. Delivery hoses shall be preserved by brushing couplings and other fittings with Type P-1 preservative and sealing openings in fittings, ends of hose, and connections with type I, grade B, pressure-sensitive tape conforming to PPP-T-60. The delivery hoses on the platform shall be removed, coiled, and tied, and stored in a weatherproof compartment.

5.1.1.4 Spray guns and nozzles. The spray guns and nozzles shall be preserved by application of Type P-10 preservative, wrapped with greaseproof barrier-material secured with pressure-sensitive tape, and place in a storage container.

5.1.1.5 Internal surfaces. All internal surfaces of the tanks (except the stainless-steel surfaces), pumps, valves, and lines of the pumping and delivery systems shall be preserved by flushing with Type P-10 preservative. Type P-1 or P-2 preservative shall be applied to the screw threads of all drain plugs and tank filler caps.

5.1.1.6 Equipment housing. All seams or openings in the equipment housing and tank enclosures shall be sealed with pressure-sensitive tape conforming to PPP-T-60.

5.1.2 Level C. The spraying unit shall be preserved and packaged to provide adequate protection from deterioration and damage to the initial receiving activity. Methods and materials shall be the minimum required by the carrier's rules or regulations. The requirements of MIL-STD-1188 shall apply.

5.2 Packing.

5.2.1 Levels A and B. Spraying units preserved and packaged as specified in 5.1.1. shall be packed in accordance with MIL-STD-281 (mobile).

5.2.2 Level C. Spraying units shall be packed in a manner to ensure safe transportation at the lowest carrier rate to the initial point of delivery. Methods and materials shall be the minimum required by the carrier's rules or regulations. The requirements of MIL-STD-1188 shall apply.

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operating under and 3.6 Performance. The spraying unit shall be capable of

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows:

SPRAYING UNIT, CLEANING-DEICING-DECONTAMINATING FLUID,
TRUCK MOUNTED AF/S32M-1

6. NOTES

6.1 Intended use. The AF/S32M-1 spraying unit is intended for use in cleaning, deicing, defrosting, and decontaminating the exterior surfaces of parked aircraft and as a work platform for maintaining, inspecting, and servicing large aircraft. The Type MA-1A gas turbine compressor is not included with the air blast system (see 3.7.13).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class of spraying unit (see 1.2)

c. When a compound system is required (see 3.7.11)

d. When an air blast system is required (see 3.7.13)

e. Registration numbers required (see 3.15.1)

f. Computations are to be furnished by the contractor to prove that the spraying unit and its components will satisfactorily withstand windloads of 7 psf. Also, that the procuring activity may accept these calculations in lieu of the windload test (see 4.6.18).

g. Point of inspection (see 4.4.1).

h. Level of preservation, packaging, and packing.

6.3 Figure 3. The following information is pertinent to figure 3:

a. Sketch A is applicable to AiResearch Mg. Co. Gas Turbine Compressor 85-20 manufactured in accordance with AiResearch Drawing 371784 and Continental Aviation and Engineering Co. Gas Turbine Compressor TC106 manufactured in accordance with Continental Drawing 568675.

b. Sketch B is applicable to AiResearch Mfg. Co. Gas Turbine Compressor 85-14 manufactured in accordance with AiResearch Drawing 370302.

c. Sketch C is applicable to AiResearch Mfg. Co. Gas Turbine Compressor 85-20 manufactured in accordance with AiResearch Drawing 370442.

6.4 Decontamination solutions consist of 40 parts by weight of by bleach and 60 parts by weight of water. Water used in preparation of this slurry should be treated with antiset agent (sugar) in the proportion of 1/2 pound to each 100 pounds of bleach (STB).

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transmission shall have a torque capacity adequate to transmit 110 perce

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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